Mathematics

1,000 Digital SAT Math Questions

Practice items matched to the published Digital SAT math blueprint.

Algebra	350 questions, about 35%
Advanced Math	350 questions, about 35%
Problem-Solving and Data Analysis	150 questions, about 15%
Geometry and Trigonometry	150 questions, about 15%

Of these 1,000 items, 810 are four-option multiple choice and 190 are student-produced responses, the kind you type as a number. On the Digital SAT itself, about 75% are multiple choice and about 25% are student-produced responses. Each item is labeled with its task family, such as slope from two points or margin of error, so the bank is not one equation type with the numbers changed. Difficulty runs from Easy through Hard.

These are original practice items, not College Board questions, and this is not an official SAT test. A calculator is appropriate throughout, as on the Digital SAT. For student-produced responses, enter an integer or a fraction in lowest terms. Do not use a mixed number. Geometry items use the standard reference facts: triangle angle sum 180°, a circle has 360°, $c = 2\pi r$, $A = \pi r^2$, rectangle area ℓw, triangle area $\frac{1}{2}bh$, Pythagorean theorem, 30°-60°-90° and 45°-45°-90° side ratios, cylinder volume $\pi r^2 h$, and right-triangle trigonometry.

The answer key begins after question 1000.

Algebra

1. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $5x + 11 = 51$?
A. 8
B. 9
C. 11
D. 7

2. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $9x + 1 = 19$?
A. 2
B. 3
C. 1
D. 18

3. Linear equations in one variable: one step · Easy · SPR
What is the solution to the equation $2x + 8 = 12$?

4. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $3x + 9 = 42$?
A. 9
B. 10
C. 12
D. 11

5. Linear equations in one variable: one step · Easy · SPR
What is the solution to the equation $9x + 7 = 34$?

6. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $3x + 9 = 15$?
A. 3
B. 2
C. 1
D. 9

7. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $3x + 10 = 49$?
A. 13
B. 10
C. 14
D. 12

8. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $6x + 3 = 39$?
A. 5
B. 3
C. 7
D. 6

9. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $5x + 11 = 71$?
A. 13
B. 12
C. 60
D. 11

10. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $6x + 12 = 54$?
A. 7
B. 8
C. 12
D. 6

11. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $4x + 2 = 18$?
A. 4
B. 5
C. 2
D. 3

12. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $5x + 5 = 50$?
A. 10
B. 9
C. 8
D. 5

13. Linear equations in one variable: one step · Easy · SPR
What is the solution to the equation $6x + 5 = 41$?

14. Linear equations in one variable: one step · Easy · MC
What is the solution to the equation $2x + 3 = 15$?
A. 5
B. 6
C. 7
D. 3

15. Linear equations in one variable: one step · Easy · SPR
What is the solution to the equation $2x + 5 = 33$?

16. Linear equations in one variable: one step · Easy · SPR
What is the solution to the equation $6x + 6 = 48$?

17. Linear equations in one variable: both sides · Medium · SPR
What value of x satisfies the equation $3x - 7 = x + 13$?

18. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $6x + 10 = 4x + 22$?
A. 6
B. -6
C. 10
D. 12

19. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $8x - 7 = 6x + 17$?
A. 18
B. 12
C. 24
D. -12

20. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $8x - 6 = x + 8$?
A. -2
B. 2
C. 3
D. 14

21. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $6x - 5 = x - 25$?
A. -3
B. 4
C. -20
D. -4

22. Linear equations in one variable: both sides · Medium · SPR
What value of x satisfies the equation $8x + 3 = 7x + 2$?

23. Linear equations in one variable: both sides · Medium · SPR
What value of x satisfies the equation $9x + 7 = 7x - 1$?

24. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $4x - 6 = 2x - 16$?
A. -10
B. -3
C. 5
D. -5

25. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $3x - 7 = x + 15$?
A. 11
B. -11
C. 12
D. 22

26. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $7x - 1 = 6x + 3$?
A. -4
B. 4
C. 10
D. 5

27. Linear equations in one variable: both sides · Medium · SPR
What value of x satisfies the equation $4x + 10 = x + 34$?

28. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $6x - 3 = 4x + 19$?
A. 22
B. 15
C. -11
D. 11

29. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $6x + 3 = x + 48$?
A. 9
B. -9
C. 10
D. 45

30. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $4x + 10 = 3x + 6$?
A. -4
B. 4
C. -1
D. -3

31. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $3x - 3 = 2x + 3$?
A. 8
B. 7
C. 6
D. -6

32. Linear equations in one variable: both sides · Medium · MC
What value of x satisfies the equation $3x + 10 = 2x + 18$?
A. -8
B. 8
C. 9
D. 10

33. Linear equations in one variable: distribution · Hard · MC
What is the solution to $3(x+5) = 4x+11$?
A. 4
B. -4
C. 7
D. 9

34. Linear equations in one variable: distribution · Hard · MC
What is the solution to $3(x+2) = 2x+12$?
A. 6
B. 8
C. -6
D. 9

35. Linear equations in one variable: distribution · Hard · MC
What is the solution to $3(x+3) = 2x+11$?
A. 5
B. 11
C. -2
D. 2

36. Linear equations in one variable: distribution · Hard · MC
What is the solution to $5(x+2) = 4x+8$?
A. 2
B. -2
C. 3
D. 0

37. Linear equations in one variable: distribution · Hard · MC
What is the solution to $2(x+3) = 3x+9$?
A. -1
B. 3
C. -3
D. 0

38. Linear equations in one variable: distribution · Hard · MC
What is the solution to $4(x+6) = 7x-3$?
A. -9
B. 9
C. 15
D. 13

39. Linear equations in one variable: distribution · Hard · MC
What is the solution to $3(x+5) = 4x+9$?
A. 11
B. 9
C. 6
D. -6

40. Linear equations in one variable: distribution · Hard · MC
What is the solution to $5(x+1)=6x+6$?
A. 1
B. 4
C. -1
D. 0

41. Linear equations in one variable: distribution · Hard · MC
What is the solution to $3(x+6)=2x+15$?
A. -3
B. 15
C. 3
D. 0

42. Linear equations in one variable: distribution · Hard · MC
What is the solution to $5(x+2)=4x+16$?
A. 8
B. 6
C. 11
D. -6

43. Linear equations in one variable: distribution · Hard · MC
What is the solution to $4(x+2)=5x-1$?
A. 13
B. 11
C. 9
D. -9

44. Linear equations in one variable: distribution · Hard · MC
What is the solution to $5(x+4)=4x+17$?
A. 1
B. 2
C. -3
D. 3

45. Linear equations in one variable: distribution · Hard · MC
What is the solution to $3(x+5)=4x+12$?
A. 8
B. -3
C. 6
D. 3

46. Linear equations in one variable: distribution · Hard · MC
What is the solution to $4(x+5)=2x+18$?
A. 4
B. 1
C. -1
D. 3

47. Linear equations in one variable: distribution · Hard · MC

What is the solution to $2(x + 6) = 3x + 6$?

A. 8

B. 6

C. -6

D. 12

48. Linear equations in one variable: distribution · Hard · MC

What is the solution to $5(x + 6) = 2x + 24$?

A. 4

B. -2

C. 3

D. 2

49. Linear equations in one variable: how many solutions · Medium · MC

How many solutions does the equation $6x + 9 = 6x + 10$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

50. Linear equations in one variable: how many solutions · Medium · MC

How many solutions does the equation $3(4x + 8) = 12x + 24$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

51. Linear equations in one variable: how many solutions · Medium · MC

How many solutions does the equation $4(6x + 3) = 24x + 12$ have?

A. Infinitely many solutions

B. Exactly one solution

C. Exactly two solutions

D. No solution

52. Linear equations in one variable: how many solutions · Medium · MC

How many solutions does the equation $2x + 3 = 2x + 4$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

53. Linear equations in one variable: how many solutions · Medium · MC

How many solutions does the equation $2x + 4 = 2x + 10$ have?

A. No solution

B. Infinitely many solutions

C. Exactly one solution

D. Exactly two solutions

54. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $2(7x + 9) = 14x + 18$ have?
A. Infinitely many solutions
B. Exactly one solution
C. Exactly two solutions
D. No solution

55. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $7x + 1 = 7x + 7$ have?
A. Exactly two solutions
B. No solution
C. Infinitely many solutions
D. Exactly one solution

56. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $3(6x + 8) = 18x + 24$ have?
A. Exactly one solution
B. Exactly two solutions
C. No solution
D. Infinitely many solutions

57. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $3x + 2 = 3x + 6$ have?
A. No solution
B. Infinitely many solutions
C. Exactly one solution
D. Exactly two solutions

58. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $3(7x + 7) = 21x + 21$ have?
A. Exactly one solution
B. Exactly two solutions
C. No solution
D. Infinitely many solutions

59. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $7x + 2 = 7x + 3$ have?
A. Exactly two solutions
B. No solution
C. Infinitely many solutions
D. Exactly one solution

60. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $4(4x + 2) = 16x + 8$ have?
A. Exactly one solution
B. Exactly two solutions
C. No solution
D. Infinitely many solutions

61. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $4(6x + 7) = 24x + 28$ have?
A. Infinitely many solutions
B. Exactly one solution
C. Exactly two solutions
D. No solution

62. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $3(7x + 8) = 21x + 24$ have?
A. No solution
B. Infinitely many solutions
C. Exactly one solution
D. Exactly two solutions

63. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $3(3x + 4) = 9x + 12$ have?
A. Infinitely many solutions
B. Exactly one solution
C. Exactly two solutions
D. No solution

64. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $2(5x + 2) = 10x + 4$ have?
A. Infinitely many solutions
B. Exactly one solution
C. Exactly two solutions
D. No solution

65. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $8x + 2 = 8x + 5$ have?
A. No solution
B. Infinitely many solutions
C. Exactly one solution
D. Exactly two solutions

66. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $3x + 3 = 3x + 9$ have?
A. No solution
B. Infinitely many solutions
C. Exactly one solution
D. Exactly two solutions

67. Linear equations in one variable: how many solutions · Medium · MC
How many solutions does the equation $7x + 5 = 7x + 11$ have?
A. Infinitely many solutions
B. Exactly one solution
C. Exactly two solutions
D. No solution

68. Linear equations in one variable: how many solutions · Medium · MC

How many solutions does the equation $6x + 7 = 6x + 12$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

69. Linear equations in two variables: slope from two points · Easy · MC

What is the slope of the line through $(5, 2)$ and $(9, -1)$?

A. $-\frac{3}{4}$

B. $\frac{3}{4}$

C. $-\frac{4}{3}$

D. -3

70. Linear equations in two variables: slope from two points · Easy · MC

What is the slope of the line through $(5, 3)$ and $(9, 1)$?

A. $-\frac{1}{2}$

B. -2

C. $\frac{1}{2}$

D. 4

71. Linear equations in two variables: slope from two points · Easy · MC

What is the slope of the line through $(5, 5)$ and $(6, 7)$?

A. -2

B. 1

C. $\frac{1}{2}$

D. 2

72. Linear equations in two variables: slope from two points · Easy · MC

What is the slope of the line through $(1, 1)$ and $(5, -1)$?

A. 4

B. $-\frac{1}{2}$

C. $\frac{1}{2}$

D. -2

73. Linear equations in two variables: slope from two points · Easy · MC

What is the slope of the line through $(2, 6)$ and $(5, 8)$?

A. $-\frac{2}{3}$

B. $\frac{2}{3}$

C. $\frac{3}{2}$

D. 2

74. Linear equations in two variables: slope from two points · Easy · MC

What is the slope of the line through $(4, 3)$ and $(7, 5)$?

A. $\frac{3}{2}$

B. $-\frac{2}{3}$

C. 2

D. $\frac{2}{3}$

75. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(5, 1)$ and $(6, 2)$?
A. 1
B. -1
C. 0
D. 2

76. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(4, 5)$ and $(6, 2)$?
A. $-\frac{3}{2}$
B. $-\frac{2}{3}$
C. $\frac{3}{2}$
D. -3

77. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(0, 5)$ and $(4, 7)$?
A. 2
B. 4
C. $\frac{1}{2}$
D. $-\frac{1}{2}$

78. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(1, 3)$ and $(5, 7)$?
A. 2
B. -1
C. 4
D. 1

79. Linear equations in two variables: slope from two points · Easy · SPR
What is the slope of the line through $(1, 4)$ and $(4, 5)$?

80. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(1, 1)$ and $(5, -2)$?
A. -3
B. $-\frac{3}{4}$
C. $-\frac{4}{3}$
D. $\frac{3}{4}$

81. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(4, 1)$ and $(8, 3)$?
A. 2
B. $\frac{1}{2}$
C. 4
D. $-\frac{1}{2}$

82. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(5, 5)$ and $(7, 2)$?
A. $-\frac{3}{2}$
B. $\frac{3}{2}$
C. -3
D. $-\frac{2}{3}$

83. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(0, 1)$ and $(3, 5)$?
A. $\frac{4}{3}$
B. 4
C. $-\frac{4}{3}$
D. $\frac{3}{4}$

84. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(2, 3)$ and $(4, 4)$?
A. 2
B. 1
C. $-\frac{1}{2}$
D. $\frac{1}{2}$

85. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(2, 0)$ and $(3, 2)$?
A. 2
B. 1
C. -2
D. $\frac{1}{2}$

86. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(2, 1)$ and $(4, 2)$?
A. 1
B. $-\frac{1}{2}$
C. $\frac{1}{2}$
D. 2

87. Linear equations in two variables: slope from two points · Easy · MC
What is the slope of the line through $(5, 3)$ and $(7, 7)$?
A. -2
B. 2
C. $\frac{1}{2}$
D. 4

88. Linear equations in two variables: slope from two points · Easy · SPR
What is the slope of the line through $(0, 1)$ and $(1, 7)$?

89. Linear functions: value on the line · Medium · MC

The line with equation $y = -3x$ passes through the point $(5, k)$. What is the value of k?

A. -15

B. -3

C. 5

D. 15

90. Linear functions: value on the line · Medium · MC

The line with equation $y = 3x + 6$ passes through the point $(5, k)$. What is the value of k?

A. 9

B. 11

C. -21

D. 21

91. Linear functions: value on the line · Medium · MC

The line with equation $y = -x - 2$ passes through the point $(2, k)$. What is the value of k?

A. -3

B. 4

C. 0

D. -4

92. Linear functions: value on the line · Medium · MC

The line with equation $y = 3x + 8$ passes through the point $(4, k)$. What is the value of k?

A. 12

B. 11

C. 20

D. 4

93. Linear functions: value on the line · Medium · MC

The line with equation $y = x - 1$ passes through the point $(4, k)$. What is the value of k?

A. 0

B. 5

C. 3

D. -3

94. Linear functions: value on the line · Medium · MC

The line with equation $y = -3x - 6$ passes through the point $(6, k)$. What is the value of k?

A. -9

B. -12

C. 0

D. -24

95. Linear functions: value on the line · Medium · MC

The line with equation $y = 5x + 1$ passes through the point $(0, k)$. What is the value of k?

A. 6

B. 1

C. -1

D. 2

96. Linear functions: value on the line · Medium · MC
The line with equation $y = 5x$ passes through the point $(0, k)$. What is the value of k?
A. -1
B. 0
C. 5
D. 1

97. Linear functions: value on the line · Medium · MC
The line with equation $y = -4x - 1$ passes through the point $(1, k)$. What is the value of k?
A. -5
B. 0
C. 5
D. -3

98. Linear functions: value on the line · Medium · MC
The line with equation $y = -3x + 1$ passes through the point $(-3, k)$. What is the value of k?
A. -10
B. 8
C. -2
D. 10

99. Linear functions: value on the line · Medium · MC
The line with equation $y = 2x - 5$ passes through the point $(6, k)$. What is the value of k?
A. 1
B. -3
C. 7
D. 17

100. Linear functions: value on the line · Medium · SPR
The line with equation $y = -2x - 6$ passes through the point $(-2, k)$. What is the value of k?

101. Linear functions: value on the line · Medium · SPR
The line with equation $y = x + 6$ passes through the point $(-3, k)$. What is the value of k?

102. Linear functions: value on the line · Medium · MC
The line with equation $y = 3x$ passes through the point $(5, k)$. What is the value of k?
A. -15
B. 5
C. 15
D. 3

103. Linear equations in two variables: parallel slope · Medium · MC
A line is parallel to the line $y = 5x - 3$. What is the slope of the parallel line?
A. 5
B. 6
C. -5
D. $-\frac{1}{5}$

104. Linear equations in two variables: parallel slope · Medium · MC
A line is parallel to the line $y = 2x$. What is the slope of the parallel line?
A. 3
B. 2
C. -2
D. $-\frac{1}{2}$

105. Linear equations in two variables: parallel slope · Medium · MC
A line is parallel to the line $y = 3x - 4$. What is the slope of the parallel line?
A. 4
B. -3
C. $-\frac{1}{3}$
D. 3

106. Linear equations in two variables: parallel slope · Medium · MC
A line is parallel to the line $y = -3x - 4$. What is the slope of the parallel line?
A. 3
B. -3
C. $\frac{1}{3}$
D. -2

107. Linear equations in two variables: parallel slope · Medium · SPR
A line is parallel to the line $y = 4x - 1$. What is the slope of the parallel line?

108. Linear equations in two variables: parallel slope · Medium · SPR
A line is parallel to the line $y = 5x + 2$. What is the slope of the parallel line?

109. Linear equations in two variables: parallel slope · Medium · MC
A line is parallel to the line $y = -4x - 5$. What is the slope of the parallel line?
A. 4
B. -4
C. -3
D. $\frac{1}{4}$

110. Linear equations in two variables: parallel slope · Medium · MC
A line is parallel to the line $y = 3x - 2$. What is the slope of the parallel line?
A. $-\frac{1}{3}$
B. -3
C. 3
D. 4

111. Linear equations in two variables: parallel slope · Medium · SPR
A line is parallel to the line $y = -3x + 6$. What is the slope of the parallel line?

112. Linear equations in two variables: parallel slope · Medium · SPR
A line is parallel to the line $y = -3x + 1$. What is the slope of the parallel line?

113. Linear equations in two variables: parallel slope · Medium · SPR
A line is parallel to the line $y = x + 6$. What is the slope of the parallel line?

114. Linear equations in two variables: parallel slope · Medium · MC

A line is parallel to the line $y = -2x + 1$. What is the slope of the parallel line?

A. -2

B. -1

C. $\frac{1}{2}$

D. 2

115. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = \frac{1}{3}x + 3$. What is the slope of the perpendicular line?

A. -3

B. 3

C. $\frac{1}{3}$

D. $-\frac{1}{3}$

116. Linear equations in two variables: perpendicular slope · Hard · SPR

A line is perpendicular to the line $y = x$. What is the slope of the perpendicular line?

117. Linear equations in two variables: perpendicular slope · Hard · SPR

A line is perpendicular to the line $y = \frac{3}{2}x + 4$. What is the slope of the perpendicular line?

118. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = \frac{2}{3}x - 1$. What is the slope of the perpendicular line?

A. $\frac{2}{3}$

B. $\frac{3}{2}$

C. $-\frac{3}{2}$

D. $-\frac{2}{3}$

119. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x + 2$. What is the slope of the perpendicular line?

A. $\frac{3}{2}$

B. $\frac{2}{3}$

C. $-\frac{2}{3}$

D. $-\frac{3}{2}$

120. Linear equations in two variables: perpendicular slope · Hard · SPR

A line is perpendicular to the line $y = 2x - 3$. What is the slope of the perpendicular line?

121. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = \frac{1}{2}x - 3$. What is the slope of the perpendicular line?

A. -2

B. $\frac{1}{2}$

C. $-\frac{1}{2}$

D. 2

122. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = 2x - 1$. What is the slope of the perpendicular line?

A. $-\frac{1}{2}$

B. 2

C. -2

D. $\frac{1}{2}$

123. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = x - 1$. What is the slope of the perpendicular line?

A. 2

B. 1

C. 0

D. -1

124. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x$. What is the slope of the perpendicular line?

A. $-\frac{2}{3}$

B. $\frac{2}{3}$

C. $\frac{3}{2}$

D. $-\frac{3}{2}$

125. Linear equations in two variables: perpendicular slope · Hard · SPR

A line is perpendicular to the line $y = 4x - 3$. What is the slope of the perpendicular line?

126. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = \frac{4}{3}x + 4$. What is the slope of the perpendicular line?

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. $-\frac{3}{4}$

D. $-\frac{4}{3}$

127. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = \frac{2}{3}x + 1$. What is the slope of the perpendicular line?

A. $\frac{3}{2}$

B. $-\frac{2}{3}$

C. $\frac{2}{3}$

D. $-\frac{3}{2}$

128. Linear equations in two variables: perpendicular slope · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x + 3$. What is the slope of the perpendicular line?

A. $-\frac{2}{3}$

B. $\frac{2}{3}$

C. $-\frac{3}{2}$

D. $\frac{3}{2}$

129. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 80 - 6x$, y is money left, in dollars and x is the number of weeks of spending. What does 6 represent?

A. The amount by which y increases for each increase of 1 in x

B. The value of y when x is 0

C. The value of x when y is 0

D. The amount by which y decreases for each increase of 1 in x

130. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 150 - 8x$, y is money left, in dollars and x is the number of weeks of spending. What does 8 represent?

A. The amount by which y decreases for each increase of 1 in x

B. The amount by which y increases for each increase of 1 in x

C. The value of y when x is 0

D. The value of x when y is 0

131. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 120 - 5x$, y is water left in a tank, in gallons and x is the number of minutes the tank has been draining. What does 5 represent?

A. The value of y when x is 0

B. The value of x when y is 0

C. The amount by which y decreases for each increase of 1 in x

D. The amount by which y increases for each increase of 1 in x

132. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 80 - 6x$, y is charge left, in percent and x is the number of hours of use. What does 6 represent?

A. The value of x when y is 0

B. The amount by which y decreases for each increase of 1 in x

C. The amount by which y increases for each increase of 1 in x

D. The value of y when x is 0

133. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 60 - 2x$, y is water left in a tank, in gallons and x is the number of minutes the tank has been draining. What does 2 represent?

A. The value of y when x is 0

B. The value of x when y is 0

C. The amount by which y decreases for each increase of 1 in x

D. The amount by which y increases for each increase of 1 in x

134. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 100 - 4x$, y is water left in a tank, in gallons and x is the number of minutes the tank has been draining. What does 4 represent?

A. The value of y when x is 0

B. The value of x when y is 0

C. The amount by which y decreases for each increase of 1 in x

D. The amount by which y increases for each increase of 1 in x

135. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 150 - 2x$, y is water left in a tank, in gallons and x is the number of minutes the tank has been draining. What does 2 represent?

A. The amount by which y increases for each increase of 1 in x

B. The value of y when x is 0

C. The value of x when y is 0

D. The amount by which y decreases for each increase of 1 in x

136. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 120 - 6x$, y is money left, in dollars and x is the number of weeks of spending. What does 6 represent?

A. The amount by which y decreases for each increase of 1 in x

B. The amount by which y increases for each increase of 1 in x

C. The value of y when x is 0

D. The value of x when y is 0

137. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 150 - 5x$, y is water left in a tank, in gallons and x is the number of minutes the tank has been draining. What does 5 represent?

A. The amount by which y increases for each increase of 1 in x

B. The value of y when x is 0

C. The value of x when y is 0

D. The amount by which y decreases for each increase of 1 in x

138. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 40 - 3x$, y is money left, in dollars and x is the number of weeks of spending. What does 3 represent?

A. The amount by which y increases for each increase of 1 in x

B. The value of y when x is 0

C. The value of x when y is 0

D. The amount by which y decreases for each increase of 1 in x

139. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 120 - 8x$, y is money left, in dollars and x is the number of weeks of spending. What does 8 represent?

A. The amount by which y increases for each increase of 1 in x

B. The value of y when x is 0

C. The value of x when y is 0

D. The amount by which y decreases for each increase of 1 in x

140. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 120 - 2x$, y is water left in a tank, in gallons and x is the number of minutes the tank has been draining. What does 2 represent?

A. The value of y when x is 0

B. The value of x when y is 0

C. The amount by which y decreases for each increase of 1 in x

D. The amount by which y increases for each increase of 1 in x

141. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 40 - 2x$, y is money left, in dollars and x is the number of weeks of spending. What does 2 represent?

A. The amount by which y increases for each increase of 1 in x

B. The value of y when x is 0

C. The value of x when y is 0

D. The amount by which y decreases for each increase of 1 in x

142. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 150 - 8x$, y is water left in a tank, in gallons and x is the number of minutes the tank has been draining. What does 8 represent?

A. The amount by which y decreases for each increase of 1 in x

B. The amount by which y increases for each increase of 1 in x

C. The value of y when x is 0

D. The value of x when y is 0

143. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 120 - 5x$, y is charge left, in percent and x is the number of hours of use. What does 5 represent?

A. The value of y when x is 0

B. The value of x when y is 0

C. The amount by which y decreases for each increase of 1 in x

D. The amount by which y increases for each increase of 1 in x

144. Linear functions: meaning of the rate · Medium · MC

In the equation $y = 80 - 3x$, y is charge left, in percent and x is the number of hours of use. What does 3 represent?

A. The amount by which y increases for each increase of 1 in x

B. The value of y when x is 0

C. The value of x when y is 0

D. The amount by which y decreases for each increase of 1 in x

145. Linear functions: meaning of the intercept · Medium · MC

In the equation $y = 120 - 3x$, y is the amount remaining and x is the number of hours. What does 120 represent?

A. The value of y when x is 0

B. The amount by which y decreases for each increase of 1 in x

C. The value of x when y is 0

D. The amount by which x increases for each increase of 1 in y

146. Linear functions: meaning of the intercept · Medium · MC

In the equation $y = 60 - 4x$, y is the amount remaining and x is the number of hours. What does 60 represent?

A. The value of y when x is 0

B. The amount by which y decreases for each increase of 1 in x

C. The value of x when y is 0

D. The amount by which x increases for each increase of 1 in y

147. Linear functions: meaning of the intercept · Medium · MC

In the equation $y = 100 - 4x$, y is the amount remaining and x is the number of hours. What does 100 represent?

A. The value of y when x is 0

B. The amount by which y decreases for each increase of 1 in x

C. The value of x when y is 0

D. The amount by which x increases for each increase of 1 in y

148. Linear functions: meaning of the intercept · Medium · MC
In the equation $y = 180 - 3x$, y is the amount remaining and x is the number of hours. What does 180 represent?
A. The amount by which x increases for each increase of 1 in y
B. The value of y when x is 0
C. The amount by which y decreases for each increase of 1 in x
D. The value of x when y is 0

149. Linear functions: meaning of the intercept · Medium · MC
In the equation $y = 100 - 2x$, y is the amount remaining and x is the number of hours. What does 100 represent?
A. The amount by which x increases for each increase of 1 in y
B. The value of y when x is 0
C. The amount by which y decreases for each increase of 1 in x
D. The value of x when y is 0

150. Linear functions: meaning of the intercept · Medium · MC
In the equation $y = 80 - 5x$, y is the amount remaining and x is the number of hours. What does 80 represent?
A. The value of x when y is 0
B. The amount by which x increases for each increase of 1 in y
C. The value of y when x is 0
D. The amount by which y decreases for each increase of 1 in x

151. Linear functions: meaning of the intercept · Medium · MC
In the equation $y = 100 - 8x$, y is the amount remaining and x is the number of hours. What does 100 represent?
A. The amount by which y decreases for each increase of 1 in x
B. The value of x when y is 0
C. The amount by which x increases for each increase of 1 in y
D. The value of y when x is 0

152. Linear functions: meaning of the intercept · Medium · MC
In the equation $y = 40 - 5x$, y is the amount remaining and x is the number of hours. What does 40 represent?
A. The amount by which y decreases for each increase of 1 in x
B. The value of x when y is 0
C. The amount by which x increases for each increase of 1 in y
D. The value of y when x is 0

153. Linear functions: meaning of the intercept · Medium · MC
In the equation $y = 180 - 4x$, y is the amount remaining and x is the number of hours. What does 180 represent?
A. The value of y when x is 0
B. The amount by which y decreases for each increase of 1 in x
C. The value of x when y is 0
D. The amount by which x increases for each increase of 1 in y

154. Linear functions: meaning of the intercept · Medium · MC
In the equation $y = 80 - 2x$, y is the amount remaining and x is the number of hours. What does 80 represent?
A. The amount by which x increases for each increase of 1 in y
B. The value of y when x is 0
C. The amount by which y decreases for each increase of 1 in x
D. The value of x when y is 0

155. Linear functions: meaning of the intercept · Medium · MC

In the equation $y = 80 - 3x$, y is the amount remaining and x is the number of hours. What does 80 represent?

A. The amount by which x increases for each increase of 1 in y

B. The value of y when x is 0

C. The amount by which y decreases for each increase of 1 in x

D. The value of x when y is 0

156. Linear functions: meaning of the intercept · Medium · MC

In the equation $y = 40 - 8x$, y is the amount remaining and x is the number of hours. What does 40 represent?

A. The value of y when x is 0

B. The amount by which y decreases for each increase of 1 in x

C. The value of x when y is 0

D. The amount by which x increases for each increase of 1 in y

157. Linear functions: meaning of the intercept · Medium · MC

In the equation $y = 80 - 4x$, y is the amount remaining and x is the number of hours. What does 80 represent?

A. The amount by which y decreases for each increase of 1 in x

B. The value of x when y is 0

C. The amount by which x increases for each increase of 1 in y

D. The value of y when x is 0

158. Linear functions: meaning of the intercept · Medium · MC

In the equation $y = 100 - 3x$, y is the amount remaining and x is the number of hours. What does 100 represent?

A. The amount by which x increases for each increase of 1 in y

B. The value of y when x is 0

C. The amount by which y decreases for each increase of 1 in x

D. The value of x when y is 0

159. Linear functions: write the model · Easy · MC

A tank contains 40 gallons and drains at 5 gallons per minute. Which equation gives g, the amount remaining after m minutes?

A. $g = 5 - 40m$

B. $g = 40 - 5m$

C. $g = 5m - 40$

D. $g = 40 + 5m$

160. Linear functions: write the model · Easy · MC

A battery is at 80 percent and loses 4 percentage points each hour. Which equation gives p, the amount remaining after h hours?

A. $p = 80 - 4h$

B. $p = 4h - 80$

C. $p = 80 + 4h$

D. $p = 4 - 80h$

161. Linear functions: write the model · Easy · MC

A tank contains 60 gallons and drains at 3 gallons per minute. Which equation gives g, the amount remaining after m minutes?

A. $g = 3m - 60$

B. $g = 60 + 3m$

C. $g = 3 - 60m$

D. $g = 60 - 3m$

162. Linear functions: write the model · Easy · MC

A tank contains 60 gallons and drains at 6 gallons per minute. Which equation gives g, the amount remaining after m minutes?

A. $g = 60 + 6m$

B. $g = 6 - 60m$

C. $g = 60 - 6m$

D. $g = 6m - 60$

163. Linear functions: write the model · Easy · MC

A tank contains 40 gallons and drains at 10 gallons per minute. Which equation gives g, the amount remaining after m minutes?

A. $g = 10m - 40$

B. $g = 40 + 10m$

C. $g = 10 - 40m$

D. $g = 40 - 10m$

164. Linear functions: write the model · Easy · MC

A student has $40 and spends $3 each week. Which equation gives d, the amount remaining after w weeks?

A. $d = 40 - 3w$

B. $d = 3w - 40$

C. $d = 40 + 3w$

D. $d = 3 - 40w$

165. Linear functions: write the model · Easy · MC

A student has $40 and spends $4 each week. Which equation gives d, the amount remaining after w weeks?

A. $d = 40 + 4w$

B. $d = 4 - 40w$

C. $d = 40 - 4w$

D. $d = 4w - 40$

166. Linear functions: write the model · Easy · MC

A battery is at 60 percent and loses 3 percentage points each hour. Which equation gives p, the amount remaining after h hours?

A. $p = 3 - 60h$

B. $p = 60 - 3h$

C. $p = 3h - 60$

D. $p = 60 + 3h$

167. Linear functions: write the model · Easy · MC

A tank contains 200 gallons and drains at 8 gallons per minute. Which equation gives g, the amount remaining after m minutes?

A. $g = 200 + 8m$

B. $g = 8 - 200m$

C. $g = 200 - 8m$

D. $g = 8m - 200$

168. Linear functions: write the model · Easy · MC

A student has \$80 and spends \$6 each week. Which equation gives d, the amount remaining after w weeks?

A. $d = 6 - 80w$

B. $d = 80 - 6w$

C. $d = 6w - 80$

D. $d = 80 + 6w$

169. Linear functions: write the model · Easy · MC

A student has \$60 and spends \$10 each week. Which equation gives d, the amount remaining after w weeks?

A. $d = 10w - 60$

B. $d = 60 + 10w$

C. $d = 10 - 60w$

D. $d = 60 - 10w$

170. Linear functions: write the model · Easy · MC

A battery is at 150 percent and loses 8 percentage points each hour. Which equation gives p, the amount remaining after h hours?

A. $p = 8h - 150$

B. $p = 150 + 8h$

C. $p = 8 - 150h$

D. $p = 150 - 8h$

171. Linear functions: write the model · Easy · MC

A student has \$80 and spends \$3 each week. Which equation gives d, the amount remaining after w weeks?

A. $d = 3 - 80w$

B. $d = 80 - 3w$

C. $d = 3w - 80$

D. $d = 80 + 3w$

172. Linear functions: write the model · Easy · MC

A student has \$40 and spends \$10 each week. Which equation gives d, the amount remaining after w weeks?

A. $d = 40 + 10w$

B. $d = 10 - 40w$

C. $d = 40 - 10w$

D. $d = 10w - 40$

173. Linear functions: write the model · Easy · MC

A battery is at 80 percent and loses 8 percentage points each hour. Which equation gives p, the amount remaining after h hours?

A. $p = 8h - 80$

B. $p = 80 + 8h$

C. $p = 8 - 80h$

D. $p = 80 - 8h$

174. Linear functions: write the model · Easy · MC

A battery is at 40 percent and loses 5 percentage points each hour. Which equation gives p, the amount remaining after h hours?

A. $p = 5h - 40$

B. $p = 40 + 5h$

C. $p = 5 - 40h$

D. $p = 40 - 5h$

175. Linear inequalities: one variable · Easy · MC

What is the solution set of $2x - 2 < 2$?

A. $x < 2$

B. $x > 2$

C. $x < 1$

D. $x \leq 2$

176. Linear inequalities: one variable · Easy · MC

What is the solution set of $3x - 2 < 28$?

A. $x < 9$

B. $x \leq 10$

C. $x < 10$

D. $x > 10$

177. Linear inequalities: one variable · Easy · MC

What is the solution set of $4x - 3 < 37$?

A. $x < 10$

B. $x > 10$

C. $x < 9$

D. $x \leq 10$

178. Linear inequalities: one variable · Easy · MC

What is the solution set of $2x - 6 < 10$?

A. $x < 8$

B. $x > 8$

C. $x < 7$

D. $x \leq 8$

179. Linear inequalities: one variable · Easy · MC
What is the solution set of $2x - 4 < 14$?
A. $x < 8$
B. $x \leq 9$
C. $x < 9$
D. $x > 9$

180. Linear inequalities: one variable · Easy · MC
What is the solution set of $5x - 4 < 26$?
A. $x \leq 6$
B. $x < 6$
C. $x > 6$
D. $x < 5$

181. Linear inequalities: one variable · Easy · MC
What is the solution set of $5x - 6 < 54$?
A. $x \leq 12$
B. $x < 12$
C. $x > 12$
D. $x < 11$

182. Linear inequalities: one variable · Easy · MC
What is the solution set of $5x - 3 < 7$?
A. $x \leq 2$
B. $x < 2$
C. $x > 2$
D. $x < 1$

183. Linear inequalities: one variable · Easy · MC
What is the solution set of $3x - 2 < 7$?
A. $x < 3$
B. $x > 3$
C. $x < 2$
D. $x \leq 3$

184. Linear inequalities: one variable · Easy · MC
What is the solution set of $3x - 6 < 30$?
A. $x < 11$
B. $x \leq 12$
C. $x < 12$
D. $x > 12$

185. Linear inequalities: one variable · Easy · MC
What is the solution set of $4x - 2 < 34$?
A. $x \leq 9$
B. $x < 9$
C. $x > 9$
D. $x < 8$

186. Linear inequalities: one variable · Easy · MC
What is the solution set of $5x - 6 < 9$?
A. $x < 3$
B. $x > 3$
C. $x < 2$
D. $x \leq 3$

187. Linear inequalities: one variable · Easy · MC
What is the solution set of $3x - 6 < 18$?
A. $x < 7$
B. $x \leq 8$
C. $x < 8$
D. $x > 8$

188. Linear inequalities: one variable · Easy · MC
What is the solution set of $3x - 3 < 6$?
A. $x < 3$
B. $x > 3$
C. $x < 2$
D. $x \leq 3$

189. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-2x + 1 < -19$?
A. $x > 11$
B. $x \geq 10$
C. $x > 10$
D. $x < 10$

190. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-5x + 2 < -28$?
A. $x > 7$
B. $x \geq 6$
C. $x > 6$
D. $x < 6$

191. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-5x + 6 < -4$?
A. $x < 2$
B. $x > 3$
C. $x \geq 2$
D. $x > 2$

192. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-2x + 5 < -9$?
A. $x > 8$
B. $x \geq 7$
C. $x > 7$
D. $x < 7$

193. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-5x + 6 < -34$?
A. $x > 8$
B. $x < 8$
C. $x > 9$
D. $x \geq 8$

194. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-2x + 8 < 2$?
A. $x < 3$
B. $x > 4$
C. $x \geq 3$
D. $x > 3$

195. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-3x + 1 < -14$?
A. $x \geq 5$
B. $x > 5$
C. $x < 5$
D. $x > 6$

196. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-2x + 8 < -10$?
A. $x \geq 9$
B. $x > 9$
C. $x < 9$
D. $x > 10$

197. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-5x + 3 < -17$?
A. $x < 4$
B. $x > 5$
C. $x \geq 4$
D. $x > 4$

198. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-3x + 6 < -3$?
A. $x > 3$
B. $x < 3$
C. $x > 4$
D. $x \geq 3$

199. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-3x + 3 < -9$?
A. $x > 4$
B. $x < 4$
C. $x > 5$
D. $x \geq 4$

200. Linear inequalities: dividing by a negative · Hard · MC
What is the solution set of $-2x + 3 < -15$?
A. $x > 9$
B. $x < 9$
C. $x > 10$
D. $x \geq 9$

201. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $1x + 2y > 5$?
A. $(0, 0)$
B. $(4, 1)$
C. $(1, 1)$
D. $(3, -1)$

202. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $4x + 1y > 9$?
A. $(-1, 5)$
B. $(1, 3)$
C. $(0, 0)$
D. $(2, 5)$

203. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $2x + 1y > 14$?
A. $(0, 0)$
B. $(5, 6)$
C. $(2, 6)$
D. $(4, 4)$

204. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $3x + 1y > 8$?
A. $(3, 2)$
B. $(0, 2)$
C. $(2, 0)$
D. $(0, 0)$

205. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $1x + 2y > 15$?
A. $(6, 5)$
B. $(3, 5)$
C. $(5, 3)$
D. $(0, 0)$

206. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $4x + 1y > 13$?
A. $(2, 3)$
B. $(0, 0)$
C. $(3, 5)$
D. $(0, 5)$

207. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $4x + 1y > 26$?
A. $(5, 4)$
B. $(0, 0)$
C. $(6, 6)$
D. $(3, 6)$

208. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $3x + 1y > 10$?
A. $(1, 1)$
B. $(3, -1)$
C. $(0, 0)$
D. $(4, 1)$

209. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $3x + 1y > 24$?
A. $(4, 6)$
B. $(6, 4)$
C. $(0, 0)$
D. $(7, 6)$

210. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $2x + 3y > 20$?
A. $(0, 0)$
B. $(5, 4)$
C. $(2, 4)$
D. $(4, 2)$

211. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $3x + 1y > 15$?
A. $(3, 4)$
B. $(0, 0)$
C. $(4, 6)$
D. $(1, 6)$

212. Linear inequalities: test a point · Medium · MC
Which point (x, y) is a solution of the inequality $1x + 2y > 7$?
A. $(1, 2)$
B. $(3, 0)$
C. $(0, 0)$
D. $(4, 2)$

213. Systems: add and subtract · Easy · SPR
The solution of the system $x + y = 3$ and $x - y = 5$ is (x, y). What is the value of x?

214. Systems: add and subtract · Easy · SPR
The solution of the system $x + y = 10$ and $x - y = -2$ is (x, y). What is the value of x?

215. Systems: add and subtract · Easy · MC
The solution of the system $x + y = 9$ and $x - y = -5$ is (x, y). What is the value of x?
A. -5
B. 7
C. 9
D. 2

216. Systems: add and subtract · Easy · MC
The solution of the system $x + y = 11$ and $x - y = -5$ is (x, y). What is the value of x?
A. 3
B. 11
C. 8
D. -5

217. Systems: add and subtract · Easy · MC
The solution of the system $x + y = 12$ and $x - y = 6$ is (x, y). What is the value of x?
A. 9
B. 3
C. 6
D. 12

218. Systems: add and subtract · Easy · SPR
The solution of the system $x + y = 3$ and $x - y = -11$ is (x, y). What is the value of x?

219. Systems: add and subtract · Easy · MC
The solution of the system $x + y = 4$ and $x - y = 8$ is (x, y). What is the value of x?
A. -2
B. 4
C. 6
D. 8

220. Systems: add and subtract · Easy · MC
The solution of the system $x + y = -1$ and $x - y = -1$ is (x, y). What is the value of x?
A. -1
B. 1
C. 0
D. -2

221. Systems: add and subtract · Easy · MC
The solution of the system $x + y = 1$ and $x - y = 9$ is (x, y). What is the value of x?
A. 1
B. -4
C. 5
D. 9

222. Systems: add and subtract · Easy · MC

The solution of the system $x + y = 4$ and $x - y = -12$ is (x, y). What is the value of x?

A. 4

B. 8

C. -4

D. -12

223. Systems: add and subtract · Easy · MC

The solution of the system $x + y = -4$ and $x - y = 2$ is (x, y). What is the value of x?

A. -4

B. -1

C. -3

D. 2

224. Systems: add and subtract · Easy · SPR

The solution of the system $x + y = 5$ and $x - y = -5$ is (x, y). What is the value of x?

225. Systems: add and subtract · Easy · MC

The solution of the system $x + y = 12$ and $x - y = 4$ is (x, y). What is the value of x?

A. 12

B. 4

C. -8

D. 8

226. Systems: add and subtract · Easy · MC

The solution of the system $x + y = 9$ and $x - y = 1$ is (x, y). What is the value of x?

A. 9

B. 5

C. 4

D. 1

227. Systems: solve for one variable · Medium · MC

What is the value of x in the solution of $3x + 2y = 23$ and $x + 2y = 9$?

A. 6

B. 8

C. 7

D. 1

228. Systems: solve for one variable · Medium · SPR

What is the value of x in the solution of $5x + 2y = 54$ and $3x + y = 31$?

229. Systems: solve for one variable · Medium · MC

What is the value of x in the solution of $2x + 3y = 13$ and $x - y = 4$?

A. 4

B. 6

C. 1

D. 5

230. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $3x + 2y = 13$ and $x + 2y = 7$?
A. 3
B. 2
C. 1
D. 5

231. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $2x + 5y = 29$ and $x - 2y = -17$?
A. 4
B. -10
C. 7
D. -3

232. Systems: solve for one variable · Medium · SPR
What is the value of x in the solution of $4x + y = -5$ and $2x - y = -1$?

233. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $5x + 2y = 2$ and $3x + y = 0$?
A. 4
B. 6
C. -2
D. -8

234. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $4x + y = 12$ and $2x - y = 6$?
A. 3
B. 12
C. 0
D. 4

235. Systems: solve for one variable · Medium · SPR
What is the value of x in the solution of $2x + 5y = 36$ and $x - 2y = 0$?

236. Systems: solve for one variable · Medium · SPR
What is the value of x in the solution of $5x + 2y = 34$ and $3x + y = 19$?

237. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $2x + 5y = -15$ and $x - 2y = 6$?
A. 0
B. -15
C. -3
D. 3

238. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $5x + 2y = 28$ and $3x + y = 16$?
A. 28
B. 4
C. 0
D. 8

239. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $2x + 5y = 36$ and $x - 2y = -9$?
A. 6
B. 9
C. -3
D. 3

240. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $5x + 2y = 37$ and $3x + y = 21$?
A. 11
B. -1
C. 6
D. 5

241. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $5x + 2y = 11$ and $3x + y = 7$?
A. 1
B. 3
C. -2
D. 5

242. Systems: solve for one variable · Medium · MC
What is the value of x in the solution of $2x + 5y = 9$ and $x - 2y = 0$?
A. 9
B. 2
C. 3
D. 1

243. Systems: value of x + y · Hard · MC
What is the value of $x + y$ in the solution of $2x + y = 2$ and $x + y = 1$?
A. 3
B. 1
C. 0
D. 2

244. Systems: value of x + y · Hard · SPR
What is the value of $x + y$ in the solution of $2x + 3y = 26$ and $x + y = 9$?

245. Systems: value of x + y · Hard · SPR
What is the value of $x + y$ in the solution of $2x + 3y = 40$ and $x + y = 16$?

246. Systems: value of x + y · Hard · SPR
What is the value of $x + y$ in the solution of $3x + y = -2$ and $x + 2y = -4$?

247. Systems: value of x + y · Hard · SPR
What is the value of $x + y$ in the solution of $4x + y = 4$ and $x + 3y = 23$?

248. Systems: value of x + y · Hard · MC
What is the value of $x + y$ in the solution of $3x + y = 5$ and $x + 2y = 5$?
A. 2
B. 3
C. -1
D. 1

249. Systems: value of x + y · Hard · SPR
What is the value of $x + y$ in the solution of $3x + y = 4$ and $x + 2y = -2$?

250. Systems: value of x + y · Hard · MC
What is the value of $x + y$ in the solution of $4x + y = 10$ and $x + 3y = 8$?
A. 4
B. 2
C. 10
D. 0

251. Systems: value of x + y · Hard · MC
What is the value of $x + y$ in the solution of $2x + 3y = 7$ and $x + y = 3$?
A. 1
B. 7
C. 2
D. 3

252. Systems: value of x + y · Hard · MC
What is the value of $x + y$ in the solution of $2x + y = 13$ and $x + y = 9$?
A. 5
B. -1
C. 9
D. 4

253. Systems: value of x + y · Hard · MC
What is the value of $x + y$ in the solution of $2x + y = 17$ and $x + y = 12$?
A. -2
B. 7
C. 5
D. 12

254. Systems: value of x + y · Hard · MC
What is the value of $x + y$ in the solution of $2x + y = 9$ and $x + y = 5$?
A. 5
B. 1
C. 3
D. 4

255. Systems: value of x + y · Hard · MC

What is the value of $x + y$ in the solution of $4x + y = 1$ and $x + 3y = 14$?

A. -6

B. 5

C. 4

D. -1

256. Systems: value of x + y · Hard · MC

What is the value of $x + y$ in the solution of $4x + y = 23$ and $x + 3y = 14$?

A. 2

B. 8

C. 3

D. 5

257. Systems: two purchases · Medium · MC

2 adult tickets and 1 student ticket cost \$43. 1 adult ticket and 3 student tickets cost \$39. What is the price, in dollars, of one adult ticket?

A. 43

B. 18

C. 7

D. 25

258. Systems: two purchases · Medium · SPR

3 adult tickets and 1 student ticket cost \$40. 1 adult ticket and 2 student tickets cost \$25. What is the price, in dollars, of one adult ticket?

259. Systems: two purchases · Medium · SPR

2 adult tickets and 3 student tickets cost \$69. 1 adult ticket and 1 student ticket cost \$29. What is the price, in dollars, of one adult ticket?

260. Systems: two purchases · Medium · SPR

3 adult tickets and 1 student ticket cost \$58. 1 adult ticket and 2 student tickets cost \$31. What is the price, in dollars, of one adult ticket?

261. Systems: two purchases · Medium · MC

2 adult tickets and 3 student tickets cost \$51. 1 adult ticket and 1 student ticket cost \$21. What is the price, in dollars, of one adult ticket?

A. 51

B. 12

C. 21

D. 9

262. Systems: two purchases · Medium · MC

4 adult tickets and 1 student ticket cost \$56. 1 adult ticket and 2 student tickets cost \$28. What is the price, in dollars, of one adult ticket?

A. 20

B. 56

C. 12

D. 8

263. Systems: two purchases · Medium · MC

4 adult tickets and 1 student ticket cost $76. 1 adult ticket and 2 student tickets cost $26. What is the price, in dollars, of one adult ticket?

A. 22

B. 4

C. 18

D. 76

264. Systems: two purchases · Medium · MC

4 adult tickets and 1 student ticket cost $59. 1 adult ticket and 2 student tickets cost $34. What is the price, in dollars, of one adult ticket?

A. 12

B. 59

C. 23

D. 11

265. Systems: two purchases · Medium · SPR

4 adult tickets and 1 student ticket cost $70. 1 adult ticket and 2 student tickets cost $35. What is the price, in dollars, of one adult ticket?

266. Systems: two purchases · Medium · MC

4 adult tickets and 1 student ticket cost $71. 1 adult ticket and 2 student tickets cost $37. What is the price, in dollars, of one adult ticket?

A. 26

B. 11

C. 15

D. 71

267. Systems: two purchases · Medium · MC

2 adult tickets and 1 student ticket cost $31. 1 adult ticket and 3 student tickets cost $28. What is the price, in dollars, of one adult ticket?

A. 31

B. 5

C. 13

D. 18

268. Systems: two purchases · Medium · MC

3 adult tickets and 1 student ticket cost $31. 1 adult ticket and 2 student tickets cost $17. What is the price, in dollars, of one adult ticket?

A. 13

B. 31

C. 9

D. 4

269. Systems: two purchases · Medium · MC

4 adult tickets and 1 student ticket cost $64. 1 adult ticket and 2 student tickets cost $30. What is the price, in dollars, of one adult ticket?

A. 8

B. 64

C. 14

D. 22

270. Systems: two purchases · Medium · MC

2 adult tickets and 1 student ticket cost \$37. 1 adult ticket and 3 student tickets cost \$36. What is the price, in dollars, of one adult ticket?

A. 7

B. 37

C. 22

D. 15

271. Systems: how many solutions · Medium · MC

How many solutions does the system $y = 3x$ and $y = 3x + 1$ have?

A. Infinitely many

B. None

C. Exactly one

D. Exactly two

272. Systems: how many solutions · Medium · MC

How many solutions does the system $y = 3x - 2$ and $y = 3x$ have?

A. None

B. Exactly one

C. Exactly two

D. Infinitely many

273. Systems: how many solutions · Medium · MC

How many solutions does the system $y = 2x - 3$ and $2y = 4x - 6$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. None

274. Systems: how many solutions · Medium · MC

How many solutions does the system $y = 4x$ and $3y = 12x$ have?

A. Exactly two

B. Infinitely many

C. None

D. Exactly one

275. Systems: how many solutions · Medium · MC

How many solutions does the system $y = 4x + 4$ and $y = 4x + 7$ have?

A. Infinitely many

B. None

C. Exactly one

D. Exactly two

276. Systems: how many solutions · Medium · MC
How many solutions does the system $y = 3x + 2$ and $2y = 6x + 4$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

277. Systems: how many solutions · Medium · MC
How many solutions does the system $y = 4x + 6$ and $y = 4x + 8$ have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. None

278. Systems: how many solutions · Medium · MC
How many solutions does the system $y = 5x - 2$ and $3y = 15x - 6$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

279. Systems: how many solutions · Medium · MC
How many solutions does the system $y = 4x + 6$ and $y = 4x + 10$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

280. Systems: how many solutions · Medium · MC
How many solutions does the system $y = 5x - 1$ and $y = 5x + 1$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

281. Systems: how many solutions · Medium · MC
How many solutions does the system $y = 4x - 1$ and $y = 4x + 1$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

282. Systems: how many solutions · Medium · MC
How many solutions does the system $y = 3x + 2$ and $y = 3x + 5$ have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. None

283. Linear functions: missing table value · Medium · MC

A linear function satisfies $f(2) = 4$ and $f(4) = 10$. What is the value of $f(6)$?

A. 4

B. 16

C. 3

D. 10

284. Linear functions: missing table value · Medium · MC

A linear function satisfies $f(1) = 18$ and $f(4) = 12$. What is the value of $f(7)$?

A. -2

B. 18

C. 6

D. 12

285. Linear functions: missing table value · Medium · MC

A linear function satisfies $f(1) = 10$ and $f(4) = 16$. What is the value of $f(7)$?

A. 10

B. 2

C. 22

D. 16

286. Linear functions: missing table value · Medium · MC

A linear function satisfies $f(2) = 8$ and $f(4) = 2$. What is the value of $f(6)$?

A. -3

B. 2

C. -4

D. 8

287. Linear functions: missing table value · Medium · SPR

A linear function satisfies $f(1) = 14$ and $f(4) = 29$. What is the value of $f(7)$?

288. Linear functions: missing table value · Medium · MC

A linear function satisfies $f(2) = 9$ and $f(4) = 3$. What is the value of $f(6)$?

A. -3

B. 9

C. -1

D. 3

289. Linear functions: missing table value · Medium · MC

A linear function satisfies $f(2) = 6$ and $f(5) = 21$. What is the value of $f(8)$?

A. 6

B. 21

C. 36

D. 5

290. Linear functions: missing table value · Medium · MC
A linear function satisfies $f(1) = 8$ and $f(4) = 23$. What is the value of $f(7)$?
A. 8
B. 23
C. 38
D. 5

291. Linear functions: missing table value · Medium · MC
A linear function satisfies $f(0) = 5$ and $f(3) = -1$. What is the value of $f(6)$?
A. -7
B. 5
C. -2
D. -1

292. Linear functions: missing table value · Medium · SPR
A linear function satisfies $f(0) = 3$ and $f(4) = 19$. What is the value of $f(8)$?

293. Linear functions: missing table value · Medium · MC
A linear function satisfies $f(0) = 5$ and $f(4) = -3$. What is the value of $f(8)$?
A. -11
B. -2
C. -3
D. 5

294. Linear functions: missing table value · Medium · MC
A linear function satisfies $f(2) = 8$ and $f(5) = 14$. What is the value of $f(8)$?
A. 8
B. 14
C. 2
D. 20

295. Linear functions: missing table value · Medium · SPR
A linear function satisfies $f(0) = 13$ and $f(4) = 1$. What is the value of $f(8)$?

296. Linear functions: missing table value · Medium · MC
A linear function satisfies $f(0) = 8$ and $f(3) = 23$. What is the value of $f(6)$?
A. 38
B. 23
C. 5
D. 8

297. Linear functions: constant rate · Easy · SPR
A car travels 260 miles in 5 hours at a constant speed. What is the speed in miles per hour?

298. Linear functions: constant rate · Easy · SPR
A car travels 208 miles in 4 hours at a constant speed. What is the speed in miles per hour?

299. Linear functions: constant rate · Easy · MC
A car travels 210 miles in 5 hours at a constant speed. What is the speed in miles per hour?
A. 210
B. 42
C. 47
D. 5

300. Linear functions: constant rate · Easy · MC
A car travels 171 miles in 3 hours at a constant speed. What is the speed in miles per hour?
A. 171
B. 3
C. 57
D. 60

301. Linear functions: constant rate · Easy · MC
A car travels 123 miles in 3 hours at a constant speed. What is the speed in miles per hour?
A. 3
B. 41
C. 44
D. 123

302. Linear functions: constant rate · Easy · SPR
A car travels 200 miles in 5 hours at a constant speed. What is the speed in miles per hour?

303. Linear functions: constant rate · Easy · SPR
A car travels 136 miles in 2 hours at a constant speed. What is the speed in miles per hour?

304. Linear functions: constant rate · Easy · MC
A car travels 240 miles in 4 hours at a constant speed. What is the speed in miles per hour?
A. 64
B. 60
C. 4
D. 240

305. Linear functions: constant rate · Easy · MC
A car travels 184 miles in 4 hours at a constant speed. What is the speed in miles per hour?
A. 184
B. 46
C. 50
D. 4

306. Linear functions: constant rate · Easy · MC
A car travels 108 miles in 2 hours at a constant speed. What is the speed in miles per hour?
A. 2
B. 108
C. 54
D. 56

307. Linear functions: constant rate · Easy · MC
A car travels 111 miles in 3 hours at a constant speed. What is the speed in miles per hour?
A. 3
B. 37
C. 111
D. 40

308. Linear functions: constant rate · Easy · MC
A car travels 116 miles in 2 hours at a constant speed. What is the speed in miles per hour?
A. 60
B. 2
C. 58
D. 116

309. Linear functions: constant rate · Easy · SPR
A car travels 165 miles in 3 hours at a constant speed. What is the speed in miles per hour?

310. Linear functions: constant rate · Easy · MC
A car travels 248 miles in 4 hours at a constant speed. What is the speed in miles per hour?
A. 248
B. 66
C. 4
D. 62

311. Linear equations: solve a formula · Easy · MC
The perimeter of a rectangle is 34 and the width is 10. What is the length?
A. 17
B. 10
C. 24
D. 7

312. Linear equations: solve a formula · Easy · SPR
The perimeter of a rectangle is 26 and the width is 6. What is the length?

313. Linear equations: solve a formula · Easy · MC
The perimeter of a rectangle is 32 and the width is 2. What is the length?
A. 14
B. 2
C. 30
D. 16

314. Linear equations: solve a formula · Easy · MC
The perimeter of a rectangle is 28 and the width is 3. What is the length?
A. 3
B. 11
C. 25
D. 14

315. Linear equations: solve a formula · Easy · MC

The perimeter of a rectangle is 40 and the width is 6. What is the length?

A. 34

B. 14

C. 20

D. 6

316. Linear equations: solve a formula · Easy · MC

The perimeter of a rectangle is 34 and the width is 6. What is the length?

A. 17

B. 6

C. 11

D. 28

317. Linear equations: solve a formula · Easy · MC

The perimeter of a rectangle is 30 and the width is 7. What is the length?

A. 8

B. 23

C. 15

D. 7

318. Linear equations: solve a formula · Easy · SPR

The perimeter of a rectangle is 30 and the width is 4. What is the length?

319. Linear equations: solve a formula · Easy · SPR

The perimeter of a rectangle is 46 and the width is 10. What is the length?

320. Linear equations: solve a formula · Easy · SPR

The perimeter of a rectangle is 16 and the width is 3. What is the length?

321. Linear equations: solve a formula · Easy · MC

The perimeter of a rectangle is 44 and the width is 6. What is the length?

A. 16

B. 22

C. 38

D. 6

322. Linear equations: solve a formula · Easy · SPR

The perimeter of a rectangle is 20 and the width is 2. What is the length?

323. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$5 each, and shipping is a flat \$6. A student can spend at most \$48. What is the greatest number of notebooks the student can buy?

A. 8

B. 9

C. 7

D. 6

324. Linear inequalities: greatest integer in a budget · Medium · SPR

Notebooks cost \$3 each, and shipping is a flat \$10. A student can spend at most \$38. What is the greatest number of notebooks the student can buy?

325. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$5 each, and shipping is a flat \$2. A student can spend at most \$54. What is the greatest number of notebooks the student can buy?

A. 10

B. 9

C. 11

D. 2

326. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$4 each, and shipping is a flat \$5. A student can spend at most \$45. What is the greatest number of notebooks the student can buy?

A. 10

B. 9

C. 11

D. 5

327. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$6 each, and shipping is a flat \$6. A student can spend at most \$31. What is the greatest number of notebooks the student can buy?

A. 6

B. 3

C. 5

D. 4

328. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$4 each, and shipping is a flat \$6. A student can spend at most \$46. What is the greatest number of notebooks the student can buy?

A. 9

B. 6

C. 11

D. 10

329. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$5 each, and shipping is a flat \$4. A student can spend at most \$49. What is the greatest number of notebooks the student can buy?

A. 4

B. 8

C. 9

D. 10

330. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$8 each, and shipping is a flat \$5. A student can spend at most \$85. What is the greatest number of notebooks the student can buy?

A. 5

B. 11

C. 10

D. 9

331. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$8 each, and shipping is a flat \$4. A student can spend at most \$45. What is the greatest number of notebooks the student can buy?

A. 5

B. 6

C. 4

D. 7

332. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$8 each, and shipping is a flat \$5. A student can spend at most \$94. What is the greatest number of notebooks the student can buy?

A. 5

B. 12

C. 10

D. 11

333. Linear inequalities: greatest integer in a budget · Medium · SPR

Notebooks cost \$5 each, and shipping is a flat \$10. A student can spend at most \$66. What is the greatest number of notebooks the student can buy?

334. Linear inequalities: greatest integer in a budget · Medium · MC

Notebooks cost \$8 each, and shipping is a flat \$4. A student can spend at most \$68. What is the greatest number of notebooks the student can buy?

A. 8

B. 9

C. 4

D. 7

335. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = -2x - 2$ and passes through $(2, -1)$. What is the y-intercept of the parallel line?

A. 1

B. -2

C. -1

D. 3

336. Linear equations in two variables: parallel line through a point · Hard · SPR

A line is parallel to $y = 4x - 4$ and passes through $(-1, 0)$. What is the y-intercept of the parallel line?

337. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = 4x + 4$ and passes through $(0, 12)$. What is the y-intercept of the parallel line?

A. 16

B. 13

C. 12

D. 4

338. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = 4x$ and passes through $(3, 12)$. What is the y-intercept of the parallel line?

A. 1

B. 4

C. 0

D. 12

339. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = 2x + 1$ and passes through $(4, 9)$. What is the y-intercept of the parallel line?

A. 9

B. 2

C. 1

D. 3

340. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = -x + 3$ and passes through $(4, 4)$. What is the y-intercept of the parallel line?

A. 4

B. 8

C. 3

D. -1

341. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = 2x + 4$ and passes through $(2, 0)$. What is the y-intercept of the parallel line?

A. -4

B. 0

C. 2

D. 4

342. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = x - 2$ and passes through $(1, 0)$. What is the y-intercept of the parallel line?

A. 1

B. -2

C. 0

D. -1

343. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = 3x + 2$ and passes through $(-2, 2)$. What is the y-intercept of the parallel line?

A. 2

B. 11

C. 3

D. 8

344. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = 4x + 5$ and passes through $(4, -1)$. What is the y-intercept of the parallel line?

A. -17

B. 4

C. 5

D. -1

345. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = -2x - 3$ and passes through $(4, -2)$. What is the y-intercept of the parallel line?

A. -3

B. -2

C. 4

D. 6

346. Linear equations in two variables: parallel line through a point · Hard · MC

A line is parallel to $y = 3x - 4$ and passes through $(0, 6)$. What is the y-intercept of the parallel line?

A. 9

B. 3

C. 6

D. -4

347. Linear equations in two variables: sign of slope and intercept · Easy · MC

Which equation has a positive slope and a positive y-intercept?

A. $y = 2x - 7$

B. $y = -2x - 7$

C. $y = 2x + 7$

D. $y = -2x + 7$

348. Linear equations in two variables: sign of slope and intercept · Easy · MC

Which equation has a positive slope and a negative y-intercept?

A. $y = 3x + 4$

B. $y = -3x + 4$

C. $y = 3x - 4$

D. $y = -3x - 4$

349. Linear equations in two variables: sign of slope and intercept · Easy · MC

Which equation has a negative slope and a positive y-intercept?

A. $y = -2x - 5$

B. $y = 2x - 5$

C. $y = -2x + 5$

D. $y = 2x + 5$

350. Linear equations in two variables: sign of slope and intercept · Easy · MC

Which equation has a negative slope and a negative y-intercept?

A. $y = -4x - 1$

B. $y = 4x - 1$

C. $y = -4x + 1$

D. $y = 4x + 1$

Advanced Math

351. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+9)(x+6)$?
A. $x^2+54x+15$
B. $x^2+15x+55$
C. $x^2-15x+54$
D. $x^2+15x+54$

352. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+1)(x+4)$?
A. x^2+5x+5
B. x^2-5x+4
C. x^2+5x+4
D. x^2+4x+5

353. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+1)(x+3)$?
A. x^2+4x+4
B. x^2-4x+3
C. x^2+4x+3
D. x^2+3x+4

354. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+8)(x+3)$?
A. $x^2+11x+24$
B. $x^2+24x+11$
C. $x^2+11x+25$
D. $x^2-11x+24$

355. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+5)(x+4)$?
A. $x^2-9x+20$
B. $x^2+9x+20$
C. $x^2+20x+9$
D. $x^2+9x+21$

356. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+3)(x+1)$?
A. x^2+3x+4
B. x^2+4x+4
C. x^2-4x+3
D. x^2+4x+3

357. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+9)(x+8)$?
A. $x^2+72x+17$
B. $x^2+17x+73$
C. $x^2-17x+72$
D. $x^2+17x+72$

358. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+1)(x+7)$?
A. x^2-8x+7
B. x^2+8x+7
C. x^2+7x+8
D. x^2+8x+8

359. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+7)(x+4)$?
A. $x^2-11x+28$
B. $x^2+11x+28$
C. $x^2+28x+11$
D. $x^2+11x+29$

360. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+1)(x+5)$?
A. x^2+5x+6
B. x^2+6x+6
C. x^2-6x+5
D. x^2+6x+5

361. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+2)(x+5)$?
A. $x^2+7x+11$
B. $x^2-7x+10$
C. $x^2+7x+10$
D. $x^2+10x+7$

362. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+8)(x+9)$?
A. $x^2+17x+72$
B. $x^2+72x+17$
C. $x^2+17x+73$
D. $x^2-17x+72$

363. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+1)(x+8)$?
A. x^2+9x+8
B. x^2+8x+9
C. x^2+9x+9
D. x^2-9x+8

364. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+8)(x+2)$?
A. $x^2+10x+17$
B. $x^2-10x+16$
C. $x^2+10x+16$
D. $x^2+16x+10$

365. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+8)(x+1)$?
A. x^2+9x+8
B. x^2+8x+9
C. x^2+9x+9
D. x^2-9x+8

366. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+6)(x+6)$?
A. $x^2+12x+37$
B. $x^2-12x+36$
C. $x^2+12x+36$
D. $x^2+36x+12$

367. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+7)(x+3)$?
A. $x^2+21x+10$
B. $x^2+10x+22$
C. $x^2-10x+21$
D. $x^2+10x+21$

368. Equivalent expressions: expand a product · Easy · MC
Which expression is equivalent to $(x+4)(x+9)$?
A. $x^2+36x+13$
B. $x^2+13x+37$
C. $x^2-13x+36$
D. $x^2+13x+36$

369. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2+8x+15$?
A. $(x-5)(x-3)$
B. $(x+5)(x+3)$
C. $(x+8)(x+15)$
D. $(x+5)(x-3)$

370. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2+10x+16$?
A. $(x+10)(x+16)$
B. $(x+2)(x-8)$
C. $(x-2)(x-8)$
D. $(x+2)(x+8)$

371. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 9x + 8$?
A. $(x + 8)(x + 1)$
B. $(x + 9)(x + 8)$
C. $(x + 8)(x - 1)$
D. $(x - 8)(x - 1)$

372. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 11x + 30$?
A. $(x + 6)(x + 5)$
B. $(x + 11)(x + 30)$
C. $(x + 6)(x - 5)$
D. $(x - 6)(x - 5)$

373. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 5x + 4$?
A. $(x + 4)(x + 1)$
B. $(x + 5)(x + 4)$
C. $(x + 4)(x - 1)$
D. $(x - 4)(x - 1)$

374. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 14x + 48$?
A. $(x + 14)(x + 48)$
B. $(x + 8)(x - 6)$
C. $(x - 8)(x - 6)$
D. $(x + 8)(x + 6)$

375. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 6x + 8$?
A. $(x + 4)(x + 2)$
B. $(x + 6)(x + 8)$
C. $(x + 4)(x - 2)$
D. $(x - 4)(x - 2)$

376. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 9x + 14$?
A. $(x + 9)(x + 14)$
B. $(x + 7)(x - 2)$
C. $(x - 7)(x - 2)$
D. $(x + 7)(x + 2)$

377. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 7x + 12$?
A. $(x + 4)(x + 3)$
B. $(x + 7)(x + 12)$
C. $(x + 4)(x - 3)$
D. $(x - 4)(x - 3)$

378. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 10x + 21$?
A. $(x + 10)(x + 21)$
B. $(x + 7)(x - 3)$
C. $(x - 7)(x - 3)$
D. $(x + 7)(x + 3)$

379. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 13x + 42$?
A. $(x - 7)(x - 6)$
B. $(x + 7)(x + 6)$
C. $(x + 13)(x + 42)$
D. $(x + 7)(x - 6)$

380. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 11x + 28$?
A. $(x + 7)(x + 4)$
B. $(x + 11)(x + 28)$
C. $(x + 7)(x - 4)$
D. $(x - 7)(x - 4)$

381. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 3x + 2$?
A. $(x + 3)(x + 2)$
B. $(x + 2)(x - 1)$
C. $(x - 2)(x - 1)$
D. $(x + 2)(x + 1)$

382. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 7x + 10$?
A. $(x + 5)(x + 2)$
B. $(x + 7)(x + 10)$
C. $(x + 5)(x - 2)$
D. $(x - 5)(x - 2)$

383. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 14x + 49$?
A. $(x - 7)(x - 7)$
B. $(x + 7)(x + 7)$
C. $(x + 14)(x + 49)$
D. $(x + 7)(x - 7)$

384. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 8x + 12$?
A. $(x - 6)(x - 2)$
B. $(x + 6)(x + 2)$
C. $(x + 8)(x + 12)$
D. $(x + 6)(x - 2)$

385. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 10x + 25$?
A. $(x + 5)(x - 5)$
B. $(x - 5)(x - 5)$
C. $(x + 5)(x + 5)$
D. $(x + 10)(x + 25)$

386. Equivalent expressions: factor a trinomial · Medium · MC
Which expression is equivalent to $x^2 + 12x + 35$?
A. $(x + 12)(x + 35)$
B. $(x + 5)(x - 7)$
C. $(x - 5)(x - 7)$
D. $(x + 5)(x + 7)$

387. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 225$?
A. $(x - 15)(x - 15)$
B. $(x + 225)(x - 1)$
C. $x(x - 15)$
D. $(x - 15)(x + 15)$

388. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 4$?
A. $(x - 2)(x - 2)$
B. $(x + 4)(x - 1)$
C. $x(x - 2)$
D. $(x - 2)(x + 2)$

389. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 64$?
A. $(x + 64)(x - 1)$
B. $x(x - 8)$
C. $(x - 8)(x + 8)$
D. $(x - 8)(x - 8)$

390. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 196$?
A. $(x + 196)(x - 1)$
B. $x(x - 14)$
C. $(x - 14)(x + 14)$
D. $(x - 14)(x - 14)$

391. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 121$?
A. $(x - 11)(x + 11)$
B. $(x - 11)(x - 11)$
C. $(x + 121)(x - 1)$
D. $x(x - 11)$

392. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 169$?
A. $(x - 13)(x - 13)$
B. $(x + 169)(x - 1)$
C. $x(x - 13)$
D. $(x - 13)(x + 13)$

393. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 16$?
A. $(x - 4)(x + 4)$
B. $(x - 4)(x - 4)$
C. $(x + 16)(x - 1)$
D. $x(x - 4)$

394. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 9$?
A. $(x - 3)(x + 3)$
B. $(x - 3)(x - 3)$
C. $(x + 9)(x - 1)$
D. $x(x - 3)$

395. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 36$?
A. $(x - 6)(x + 6)$
B. $(x - 6)(x - 6)$
C. $(x + 36)(x - 1)$
D. $x(x - 6)$

396. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 25$?
A. $(x - 5)(x - 5)$
B. $(x + 25)(x - 1)$
C. $x(x - 5)$
D. $(x - 5)(x + 5)$

397. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 49$?
A. $(x + 49)(x - 1)$
B. $x(x - 7)$
C. $(x - 7)(x + 7)$
D. $(x - 7)(x - 7)$

398. Equivalent expressions: difference of squares · Medium · MC
Which expression is equivalent to $x^2 - 144$?
A. $(x - 12)(x + 12)$
B. $(x - 12)(x - 12)$
C. $(x + 144)(x - 1)$
D. $x(x - 12)$

399. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^4 \cdot x^8$ for $x \neq 0$?
A. $32x$
B. x^{12}
C. x^{32}
D. x^4

400. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^8 \cdot x^4$ for $x \neq 0$?
A. x^4
B. $32x$
C. x^{12}
D. x^{32}

401. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^9 \cdot x^8$ for $x \neq 0$?
A. x^{72}
B. x^1
C. $72x$
D. x^{17}

402. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^5 \cdot x^4$ for $x \neq 0$?
A. x^9
B. x^{20}
C. x^1
D. $20x$

403. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^2 \cdot x^6$ for $x \neq 0$?
A. x^4
B. $12x$
C. x^8
D. x^{12}

404. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^3 \cdot x^4$ for $x \neq 0$?
A. x^{12}
B. x^1
C. $12x$
D. x^7

405. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^3 \cdot x^5$ for $x \neq 0$?
A. x^2
B. $15x$
C. x^8
D. x^{15}

406. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^2 \cdot x^8$ for $x \neq 0$?
A. $16x$
B. x^{10}
C. x^{16}
D. x^6

407. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^5 \cdot x^3$ for $x \neq 0$?
A. x^8
B. x^{15}
C. x^2
D. $15x$

408. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^9 \cdot x^5$ for $x \neq 0$?
A. x^4
B. $45x$
C. x^{14}
D. x^{45}

409. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^5 \cdot x^8$ for $x \neq 0$?
A. $40x$
B. x^{13}
C. x^{40}
D. x^3

410. Equivalent expressions: product of powers · Easy · MC
Which expression is equivalent to $x^9 \cdot x^2$ for $x \neq 0$?
A. x^7
B. $18x$
C. x^{11}
D. x^{18}

411. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^4)^3$?
A. x^{12}
B. x^7
C. x^4
D. $12x$

412. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^5)^4$?
A. x^5
B. $20x$
C. x^{20}
D. x^9

413. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^8)^4$?
A. x^{32}
B. x^{12}
C. x^8
D. $32x$

414. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^3)^2$?
A. x^5
B. x^3
C. $6x$
D. x^6

415. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^3)^3$?
A. x^9
B. x^6
C. x^3
D. $9x$

416. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^6)^5$?
A. $30x$
B. x^{30}
C. x^{11}
D. x^6

417. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^4)^6$?
A. x^{24}
B. x^{10}
C. x^4
D. $24x$

418. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^7)^5$?
A. x^7
B. $35x$
C. x^{35}
D. x^{12}

419. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^7)^2$?
A. x^7
B. $14x$
C. x^{14}
D. x^9

420. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^6)^2$?
A. $12x$
B. x^{12}
C. x^8
D. x^6

421. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^8)^2$?
A. x^8
B. $16x$
C. x^{16}
D. x^{10}

422. Equivalent expressions: power of a power · Medium · MC
Which expression is equivalent to $(x^7)^3$?
A. x^{10}
B. x^7
C. $21x$
D. x^{21}

423. Nonlinear equations: sum of solutions · Easy · SPR
What is the sum of the solutions of $x^2 - 9x + 8 = 0$?

424. Nonlinear equations: sum of solutions · Easy · SPR
What is the sum of the solutions of $x^2 + 5x + 4 = 0$?

425. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 8x = 0$?
A. 9
B. 0
C. 7
D. 8

426. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 6x + 8 = 0$?
A. 6
B. 8
C. 4
D. 2

427. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 16 = 0$?
A. 4
B. -4
C. -16
D. 0

428. Nonlinear equations: sum of solutions · Easy · SPR
What is the sum of the solutions of $x^2 - 14x + 48 = 0$?

429. Nonlinear equations: sum of solutions · Easy · SPR
What is the sum of the solutions of $x^2 - 12x + 35 = 0$?

430. Nonlinear equations: sum of solutions · Easy · SPR
What is the sum of the solutions of $x^2 - 7x + 10 = 0$?

431. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 1x = 0$?
A. 1
B. 3
C. 0
D. 2

432. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 8x + 12 = 0$?
A. 6
B. 12
C. 2
D. 8

433. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 + 1x - 2 = 0$?
A. 1
B. -2
C. 3
D. -1

434. Nonlinear equations: sum of solutions · Easy · SPR
What is the sum of the solutions of $x^2 - 4x - 32 = 0$?

435. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 9x + 18 = 0$?
A. 6
B. 18
C. 9
D. 3

436. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 5x = 0$?
A. 5
B. 4
C. 6
D. 0

437. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 15x + 56 = 0$?
A. 15
B. 8
C. 56
D. 7

438. Nonlinear equations: sum of solutions · Easy · SPR
What is the sum of the solutions of $x^2 - 4x - 21 = 0$?

439. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 + 2x - 3 = 0$?
A. -2
B. 4
C. -3
D. 1

440. Nonlinear equations: sum of solutions · Easy · MC
What is the sum of the solutions of $x^2 - 5x - 36 = 0$?
A. -36
B. 9
C. -4
D. 5

441. Nonlinear equations: product of solutions · Medium · MC
What is the product of the solutions of $x^2 - 7x - 18 = 0$?
A. -2
B. 9
C. -18
D. 7

442. Nonlinear equations: product of solutions · Medium · MC
What is the product of the solutions of $x^2 + 2x - 15 = 0$?
A. -15
B. -5
C. -2
D. 3

443. Nonlinear equations: product of solutions · Medium · MC
What is the product of the solutions of $x^2 + 4x + 3 = 0$?
A. -3
B. -4
C. 3
D. -1

444. Nonlinear equations: product of solutions · Medium · SPR
What is the product of the solutions of $x^2 + 1x - 12 = 0$?

445. Nonlinear equations: product of solutions · Medium · MC
What is the product of the solutions of $x^2 - 9 = 0$?
A. 3
B. -9
C. 0
D. -3

446. Nonlinear equations: product of solutions · Medium · MC
What is the product of the solutions of $x^2 + 4x = 0$?
A. 0
B. -4
C. 4
D. 1

447. Nonlinear equations: product of solutions · Medium · MC
What is the product of the solutions of $x^2 - 3x - 18 = 0$?
A. -18
B. 3
C. -3
D. 6

448. Nonlinear equations: product of solutions · Medium · SPR
What is the product of the solutions of $x^2 - 2x - 24 = 0$?

449. Nonlinear equations: product of solutions · Medium · MC
What is the product of the solutions of $x^2 - 10x + 9 = 0$?
A. 8
B. 1
C. 9
D. 10

450. Nonlinear equations: product of solutions · Medium · SPR
What is the product of the solutions of $x^2 - 11x + 24 = 0$?

451. Nonlinear equations: product of solutions · Medium · MC
What is the product of the solutions of $x^2 - 8x = 0$?
A. -1
B. 8
C. 1
D. 0

452. Nonlinear equations: product of solutions · Medium · SPR
What is the product of the solutions of $x^2 - 19x + 88 = 0$?

453. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 1x - 12 = 0$?
A. -12
B. 4
C. 1
D. -3

454. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 6x + 5 = 0$?
A. 6
B. 1
C. 5
D. 4

455. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 9x + 20 = 0$?
A. 4
B. 20
C. 9
D. 5

456. Nonlinear equations: greater solution · Medium · SPR
What is the greater solution of $x^2 - 8x = 0$?

457. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 15x + 56 = 0$?
A. 56
B. 15
C. 7
D. 8

458. Nonlinear equations: greater solution · Medium · SPR
What is the greater solution of $x^2 + 2x - 8 = 0$?

459. Nonlinear equations: greater solution · Medium · SPR
What is the greater solution of $x^2 - 7x - 8 = 0$?

460. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 3x = 0$?
A. 0
B. 2
C. 4
D. 3

461. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 11x + 28 = 0$?
A. 11
B. 4
C. 7
D. 28

462. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 3x + 2 = 0$?
A. 1
B. 2
C. 3
D. 4

463. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 1x - 2 = 0$?
A. -1
B. 2
C. 1
D. -2

464. Nonlinear equations: greater solution · Medium · SPR
What is the greater solution of $x^2 - 4x - 21 = 0$?

465. Nonlinear equations: greater solution · Medium · SPR
What is the greater solution of $x^2 - 12x + 35 = 0$?

466. Nonlinear equations: greater solution · Medium · MC
What is the greater solution of $x^2 - 7x + 10 = 0$?
A. 5
B. 10
C. 7
D. 2

467. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 14x + 55$ is defined for all real x. What is the minimum value of f?
A. -7
B. 6
C. 7
D. 55

468. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 8x + 16$ is defined for all real x. What is the minimum value of f?
A. 16
B. -4
C. 4
D. 0

469. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 10x + 33$ is defined for all real x. What is the minimum value of f?
A. -5
B. 5
C. 8
D. 33

470. Nonlinear functions: minimum value · Hard · SPR
The function $f(x) = x^2 - 2x + -4$ is defined for all real x. What is the minimum value of f?

471. Nonlinear functions: minimum value · Hard · SPR
The function $f(x) = x^2 - 16x + 73$ is defined for all real x. What is the minimum value of f?

472. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 12x + 43$ is defined for all real x. What is the minimum value of f?
A. 6
B. -6
C. 7
D. 43

473. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 6x + 12$ is defined for all real x. What is the minimum value of f?
A. -3
B. 3
C. 6
D. 12

474. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 12x + 30$ is defined for all real x. What is the minimum value of f?
A. 12
B. -6
C. 30
D. 6

475. Nonlinear functions: minimum value · Hard · SPR
The function $f(x) = x^2 - 8x + 15$ is defined for all real x. What is the minimum value of f?

476. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 16x + 68$ is defined for all real x. What is the minimum value of f?
A. 68
B. -8
C. 8
D. 4

477. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 4x + 10$ is defined for all real x. What is the minimum value of f?
A. 6
B. -2
C. 10
D. 2

478. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 16x + 60$ is defined for all real x. What is the minimum value of f?
A. -4
B. 60
C. 8
D. -8

479. Nonlinear functions: minimum value · Hard · SPR
The function $f(x) = x^2 - 4x + 9$ is defined for all real x. What is the minimum value of f?

480. Nonlinear functions: minimum value · Hard · MC
The function $f(x) = x^2 - 10x + 19$ is defined for all real x. What is the minimum value of f?
A. -5
B. 5
C. -6
D. 19

481. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 10x + 31$ is defined for all real x. What is the x-coordinate of the vertex?
A. 6
B. -5
C. 10
D. 5

482. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 16x + 62$ is defined for all real x. What is the x-coordinate of the vertex?
A. 8
B. 16
C. -2
D. -8

483. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 12x + 32$ is defined for all real x. What is the x-coordinate of the vertex?
A. -6
B. -4
C. 6
D. 12

484. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 4x + 7$ is defined for all real x. What is the x-coordinate of the vertex?
A. -2
B. 4
C. 3
D. 2

485. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 16x + 66$ is defined for all real x. What is the x-coordinate of the vertex?
A. 2
B. -8
C. 8
D. 16

486. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 2x + 7$ is defined for all real x. What is the x-coordinate of the vertex?
A. 6
B. -1
C. 1
D. 2

487. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 6x + 14$ is defined for all real x. What is the x-coordinate of the vertex?
A. 3
B. -3
C. 5
D. 6

488. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 6x + 7$ is defined for all real x. What is the x-coordinate of the vertex?
A. 3
B. -2
C. 6
D. -3

489. Nonlinear functions: x-coordinate of the vertex · Medium · SPR
The function $f(x) = x^2 - 18x + 85$ is defined for all real x. What is the x-coordinate of the vertex?

490. Nonlinear functions: x-coordinate of the vertex · Medium · SPR
The function $f(x) = x^2 - 8x + 11$ is defined for all real x. What is the x-coordinate of the vertex?

491. Nonlinear functions: x-coordinate of the vertex · Medium · SPR
The function $f(x) = x^2 - 18x + 79$ is defined for all real x. What is the x-coordinate of the vertex?

492. Nonlinear functions: x-coordinate of the vertex · Medium · MC
The function $f(x) = x^2 - 8x + 20$ is defined for all real x. What is the x-coordinate of the vertex?
A. 20
B. 8
C. 4
D. -4

493. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 2x + 6 = 0$ have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. None

494. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 4x + 8 = 0$ have?
A. Infinitely many
B. None
C. Exactly one
D. Exactly two

495. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 - 19x + 88 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

496. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 12x + 36 = 0$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

497. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 2 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

498. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 - 10x + 25 = 0$ have?
A. Infinitely many
B. None
C. Exactly one
D. Exactly two

499. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 1x - 2 = 0$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

500. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 8x + 16 = 0$ have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. None

501. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 2x + 1 = 0$ have?
A. Infinitely many
B. None
C. Exactly one
D. Exactly two

502. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 4x + 10 = 0$ have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. None

503. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 + 5 = 0$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

504. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 - 9x + 20 = 0$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

505. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 - 1 = 0$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

506. Nonlinear equations: how many real solutions · Medium · MC
How many distinct real solutions does $x^2 - 17x + 72 = 0$ have?
A. Infinitely many
B. None
C. Exactly one
D. Exactly two

507. Nonlinear functions: evaluate an exponential · Easy · SPR
If $f(t) = 6 \cdot 3^t$, what is the value of $f(2)$?

508. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 6 \cdot 4^t$, what is the value of $f(2)$?
A. 48
B. 16
C. 96
D. 22

509. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 9 \cdot 4^t$, what is the value of $f(2)$?
A. 72
B. 16
C. 144
D. 25

510. Nonlinear functions: evaluate an exponential · Easy · SPR
If $f(t) = 6 \cdot 4^t$, what is the value of $f(4)$?

511. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 6 \cdot 3^t$, what is the value of $f(3)$?
A. 162
B. 33
C. 54
D. 27

512. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 2 \cdot 4^t$, what is the value of $f(3)$?
A. 66
B. 24
C. 128
D. 64

513. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 9 \cdot 4^t$, what is the value of $f(3)$?
A. 108
B. 576
C. 64
D. 73

514. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 3 \cdot 2^t$, what is the value of $f(2)$?
A. 12
B. 4
C. 7
D. 8

515. Nonlinear functions: evaluate an exponential · Easy · SPR
If $f(t) = 7 \cdot 2^t$, what is the value of $f(4)$?

516. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 3 \cdot 3^t$, what is the value of $f(3)$?
A. 30
B. 27
C. 81
D. 12

517. Nonlinear functions: evaluate an exponential · Easy · SPR
If $f(t) = 5 \cdot 4^t$, what is the value of $f(4)$?

518. Nonlinear functions: evaluate an exponential · Easy · SPR
If $f(t) = 6 \cdot 2^t$, what is the value of $f(2)$?

519. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 8 \cdot 3^t$, what is the value of $f(2)$?
A. 9
B. 72
C. 48
D. 17

520. Nonlinear functions: evaluate an exponential · Easy · MC
If $f(t) = 3 \cdot 2^t$, what is the value of $f(4)$?
A. 48
B. 24
C. 16
D. 19

521. Nonlinear functions: initial value of an exponential · Hard · MC
A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 224$, what is the value of a?
A. 18
B. 56
C. 16
D. 14

522. Nonlinear functions: initial value of an exponential · Hard · MC
A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 112$, what is the value of a?
A. 16
B. 28
C. 7
D. 11

523. Nonlinear functions: initial value of an exponential · Hard · MC
A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 48$, what is the value of a?
A. 12
B. 16
C. 7
D. 3

524. Nonlinear functions: initial value of an exponential · Hard · MC
A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 192$, what is the value of a?
A. 24
B. 16
C. 48
D. 12

525. Nonlinear functions: initial value of an exponential · Hard · SPR
A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 96$, what is the value of a?

526. Nonlinear functions: initial value of an exponential · Hard · SPR
A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 108$, what is the value of a?

527. Nonlinear functions: initial value of an exponential · Hard · MC
A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 80$, what is the value of a?
A. 20
B. 16
C. 9
D. 5

528. Nonlinear functions: initial value of an exponential · Hard · MC
A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 27$, what is the value of a?
A. 6
B. 4
C. 3
D. 9

529. Nonlinear functions: initial value of an exponential · Hard · MC

A quantity grows by the model $P(t) = a \cdot 2^t$. If $P(2) = 44$, what is the value of a?

A. 4

B. 11

C. 22

D. 13

530. Nonlinear functions: initial value of an exponential · Hard · MC

A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 160$, what is the value of a?

A. 40

B. 10

C. 14

D. 16

531. Nonlinear functions: initial value of an exponential · Hard · MC

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 135$, what is the value of a?

A. 45

B. 15

C. 18

D. 9

532. Nonlinear functions: initial value of an exponential · Hard · MC

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 117$, what is the value of a?

A. 13

B. 9

C. 16

D. 39

533. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 50 and is multiplied by 2 each year. Which expression gives the population after t years?

A. $2 \cdot 50^t$

B. $50 \cdot 2^t$

C. $50 \cdot t^2$

D. $52t$

534. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 40 and is multiplied by 3 each year. Which expression gives the population after t years?

A. $3 \cdot 40^t$

B. $40 \cdot 3^t$

C. $40 \cdot t^3$

D. $43t$

535. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 20 and is multiplied by 2 each year. Which expression gives the population after t years?

A. $20 \cdot t^2$

B. $22t$

C. $2 \cdot 20^t$

D. $20 \cdot 2^t$

536. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 50 and is multiplied by 3 each year. Which expression gives the population after t years?

A. $53t$

B. $3 \cdot 50^t$

C. $50 \cdot 3^t$

D. $50 \cdot t^3$

537. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 40 and is multiplied by 4 each year. Which expression gives the population after t years?

A. $44t$

B. $4 \cdot 40^t$

C. $40 \cdot 4^t$

D. $40 \cdot t^4$

538. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 80 and is multiplied by 3 each year. Which expression gives the population after t years?

A. $80 \cdot t^3$

B. $83t$

C. $3 \cdot 80^t$

D. $80 \cdot 3^t$

539. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 20 and is multiplied by 4 each year. Which expression gives the population after t years?

A. $4 \cdot 20^t$

B. $20 \cdot 4^t$

C. $20 \cdot t^4$

D. $24t$

540. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 40 and is multiplied by 2 each year. Which expression gives the population after t years?

A. $2 \cdot 40^t$

B. $40 \cdot 2^t$

C. $40 \cdot t^2$

D. $42t$

541. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 100 and is multiplied by 2 each year. Which expression gives the population after t years?

A. $100 \cdot t^2$

B. $102t$

C. $2 \cdot 100^t$

D. $100 \cdot 2^t$

542. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 80 and is multiplied by 2 each year. Which expression gives the population after t years?

A. $82t$

B. $2 \cdot 80^t$

C. $80 \cdot 2^t$

D. $80 \cdot t^2$

543. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 80 and is multiplied by 4 each year. Which expression gives the population after t years?

A. $80 \cdot 4^t$

B. $80 \cdot t^4$

C. $84t$

D. $4 \cdot 80^t$

544. Nonlinear functions: choose the exponential model · Medium · MC

A population starts at 20 and is multiplied by 3 each year. Which expression gives the population after t years?

A. $20 \cdot t^3$

B. $23t$

C. $3 \cdot 20^t$

D. $20 \cdot 3^t$

545. Nonlinear functions: half-life · Medium · MC

A sample starts at 120 grams and is halved each year. How many grams remain after 2 years?

A. 240

B. 30

C. 60

D. 118

546. Nonlinear functions: half-life · Medium · MC

A sample starts at 320 grams and is halved each year. How many grams remain after 3 years?

A. 960

B. 40

C. 317

D. 160

547. Nonlinear functions: half-life · Medium · SPR

A sample starts at 120 grams and is halved each year. How many grams remain after 3 years?

548. Nonlinear functions: half-life · Medium · MC

A sample starts at 240 grams and is halved each year. How many grams remain after 2 years?

A. 480

B. 60

C. 120

D. 238

549. Nonlinear functions: half-life · Medium · SPR

A sample starts at 160 grams and is halved each year. How many grams remain after 3 years?

550. Nonlinear functions: half-life · Medium · SPR

A sample starts at 80 grams and is halved each year. How many grams remain after 3 years?

551. Nonlinear functions: half-life · Medium · SPR

A sample starts at 80 grams and is halved each year. How many grams remain after 2 years?

552. Nonlinear functions: half-life · Medium · MC
A sample starts at 240 grams and is halved each year. How many grams remain after 3 years?
A. 237
B. 720
C. 120
D. 30

553. Nonlinear functions: evaluate f(x) · Easy · SPR
If $f(x) = 6x - 4$, what is the value of $f(2)$?

554. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 5x - 4$, what is the value of $f(-2)$?
A. -1
B. -5
C. -14
D. -6

555. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 3x + 4$, what is the value of $f(3)$?
A. 10
B. 12
C. 5
D. 13

556. Nonlinear functions: evaluate f(x) · Easy · SPR
If $f(x) = 4x - 2$, what is the value of $f(2)$?

557. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 5x + 2$, what is the value of $f(-3)$?
A. -10
B. -17
C. 4
D. -13

558. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 6x - 2$, what is the value of $f(-1)$?
A. 0
B. 3
C. -4
D. -8

559. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 6x + 7$, what is the value of $f(5)$?
A. 18
B. 23
C. 36
D. 37

560. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 4x + 6$, what is the value of $f(-1)$?
A. 9
B. -10
C. 2
D. 0

561. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 3x + 1$, what is the value of $f(-3)$?
A. 1
B. -10
C. -6
D. -8

562. Nonlinear functions: evaluate f(x) · Easy · SPR
If $f(x) = 2x + 6$, what is the value of $f(1)$?

563. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 5x + 1$, what is the value of $f(0)$?
A. -1
B. 5
C. 6
D. 1

564. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 3x - 4$, what is the value of $f(6)$?
A. 21
B. 14
C. 22
D. 5

565. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 4x + 4$, what is the value of $f(3)$?
A. 8
B. 17
C. 16
D. 11

566. Nonlinear functions: evaluate f(x) · Easy · MC
If $f(x) = 4x - 5$, what is the value of $f(7)$?
A. 6
B. 32
C. 33
D. 23

567. Nonlinear functions: composition · Medium · SPR
If $f(x) = 3x - 3$ and $g(x) = 3x - 3$, what is the value of $f(g(3))$?

568. Nonlinear functions: composition · Medium · MC
If $f(x) = 3x + 3$ and $g(x) = 2x + 4$, what is the value of $f(g(1))$?
A. 6
B. 12
C. 21
D. 22

569. Nonlinear functions: composition · Medium · MC
If $f(x) = 4x - 1$ and $g(x) = 2x$, what is the value of $f(g(1))$?
A. 6
B. 7
C. 5
D. 8

570. Nonlinear functions: composition · Medium · MC
If $f(x) = 3x + 3$ and $g(x) = 2x - 1$, what is the value of $f(g(3))$?
A. 19
B. 20
C. 17
D. 18

571. Nonlinear functions: composition · Medium · MC
If $f(x) = 2x - 4$ and $g(x) = 4x + 3$, what is the value of $f(g(0))$?
A. 3
B. -1
C. 0
D. 2

572. Nonlinear functions: composition · Medium · MC
If $f(x) = 5x + 2$ and $g(x) = 2x$, what is the value of $f(g(2))$?
A. 23
B. 20
C. 16
D. 22

573. Nonlinear functions: composition · Medium · MC
If $f(x) = 2x - 2$ and $g(x) = 2x - 2$, what is the value of $f(g(-2))$?
A. -13
B. -14
C. -8
D. -12

574. Nonlinear functions: composition · Medium · SPR
If $f(x) = 2x + 6$ and $g(x) = x + 4$, what is the value of $f(g(1))$?

575. Nonlinear functions: composition · Medium · MC
If $f(x) = 2x + 1$ and $g(x) = 2x + 4$, what is the value of $f(g(2))$?
A. 8
B. 18
C. 17
D. 13

576. Nonlinear functions: composition · Medium · MC
If $f(x) = 4x - 4$ and $g(x) = 3x + 2$, what is the value of $f(g(2))$?
A. 24
B. 29
C. 12
D. 28

577. Nonlinear functions: composition · Medium · MC
If $f(x) = 2x - 3$ and $g(x) = 2x - 2$, what is the value of $f(g(1))$?
A. -3
B. 4
C. -1
D. -2

578. Nonlinear functions: composition · Medium · SPR
If $f(x) = 3x$ and $g(x) = 2x + 1$, what is the value of $f(g(2))$?

579. Nonlinear functions: composition · Medium · SPR
If $f(x) = 4x + 1$ and $g(x) = 2x + 2$, what is the value of $f(g(5))$?

580. Nonlinear functions: composition · Medium · SPR
If $f(x) = 5x$ and $g(x) = 4x$, what is the value of $f(g(-2))$?

581. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 3)$. If $f(4) = -4$, what is the value of $g(7)$?
A. -1
B. -4
C. -7
D. 4

582. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 4)$. If $f(2) = 1$, what is the value of $g(6)$?
A. 5
B. 1
C. -3
D. 2

583. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 1)$. If $f(6) = -5$, what is the value of $g(7)$?
A. -6
B. -4
C. -5
D. 6

584. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 3)$. If $f(2) = 4$, what is the value of $g(5)$?
A. 1
B. 7
C. 2
D. 4

585. Nonlinear functions: horizontal shift · Hard · SPR
The function g is defined by $g(x) = f(x - 4)$. If $f(3) = -3$, what is the value of $g(7)$?

586. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 4)$. If $f(5) = 0$, what is the value of $g(9)$?
A. 5
B. 4
C. -4
D. 0

587. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 4)$. If $f(2) = 3$, what is the value of $g(6)$?
A. 3
B. 7
C. -1
D. 2

588. Nonlinear functions: horizontal shift · Hard · SPR
The function g is defined by $g(x) = f(x - 3)$. If $f(4) = 5$, what is the value of $g(7)$?

589. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 1)$. If $f(2) = 14$, what is the value of $g(3)$?
A. 14
B. 15
C. 2
D. 13

590. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 3)$. If $f(5) = 3$, what is the value of $g(8)$?
A. 5
B. 3
C. 6
D. 0

591. Nonlinear functions: horizontal shift · Hard · MC
The function g is defined by $g(x) = f(x - 2)$. If $f(6) = 0$, what is the value of $g(8)$?
A. 6
B. 0
C. 2
D. -2

592. Nonlinear functions: horizontal shift · Hard · MC

The function g is defined by $g(x) = f(x - 4)$. If $f(4) = -4$, what is the value of $g(8)$?

A. 4

B. -8

C. -4

D. 0

593. Nonlinear equations: radical · Medium · MC

What is the solution of $\sqrt{x + 7} = 3$?

A. 16

B. 9

C. 10

D. 2

594. Nonlinear equations: radical · Medium · MC

What is the solution of $\sqrt{x + 1} = 2$?

A. 1

B. 5

C. 4

D. 3

595. Nonlinear equations: radical · Medium · MC

What is the solution of $\sqrt{x + 5} = 8$?

A. 13

B. 59

C. 69

D. 64

596. Nonlinear equations: radical · Medium · MC

What is the solution of $\sqrt{x + 8} = 9$?

A. 89

B. 73

C. 17

D. 81

597. Nonlinear equations: radical · Medium · MC

What is the solution of $\sqrt{x + 7} = 5$?

A. 32

B. 25

C. 18

D. 12

598. Nonlinear equations: radical · Medium · MC

What is the solution of $\sqrt{x + 2} = 7$?

A. 47

B. 9

C. 51

D. 49

599. Nonlinear equations: radical · Medium · SPR
What is the solution of $\sqrt{x+2} = 2$?

600. Nonlinear equations: radical · Medium · SPR
What is the solution of $\sqrt{x+10} = 9$?

601. Nonlinear equations: radical · Medium · MC
What is the solution of $\sqrt{x+8} = 7$?
A. 15
B. 41
C. 49
D. 57

602. Nonlinear equations: radical · Medium · SPR
What is the solution of $\sqrt{x+12} = 6$?

603. Nonlinear equations: radical · Medium · MC
What is the solution of $\sqrt{x+12} = 4$?
A. 16
B. 28
C. 12
D. 4

604. Nonlinear equations: radical · Medium · MC
What is the solution of $\sqrt{x+12} = 8$?
A. 20
B. 64
C. 52
D. 76

605. Nonlinear equations: radical · Medium · MC
What is the solution of $\sqrt{x+5} = 3$?
A. 9
B. 8
C. 14
D. 4

606. Nonlinear equations: radical · Medium · MC
What is the solution of $\sqrt{x+1} = 5$?
A. 6
B. 26
C. 24
D. 25

607. Nonlinear equations: rational · Hard · MC
What value of x satisfies $\dfrac{x+25}{x+1} = 4$, given that the expression is defined?
A. -1
B. 11
C. 32
D. 7

608. Nonlinear equations: rational · Hard · SPR

What value of x satisfies $\dfrac{x+28}{x+1} = 4$, given that the expression is defined?

609. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x+45}{x+6} = 4$, given that the expression is defined?

A. 7

B. 11

C. -6

D. 52

610. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x+1}{x+2} = 2$, given that the expression is defined?

A. -1

B. -3

C. 1

D. -2

611. Nonlinear equations: rational · Hard · SPR

What value of x satisfies $\dfrac{x+9}{x+1} = 2$, given that the expression is defined?

612. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x+4}{x+3} = 2$, given that the expression is defined?

A. -3

B. 2

C. 0

D. -2

613. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x+16}{x+5} = 2$, given that the expression is defined?

A. 8

B. 6

C. 22

D. -5

614. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x-2}{x+2} = 3$, given that the expression is defined?

A. -6

B. -1

C. -4

D. -2

615. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x+16}{x+2}=3$, given that the expression is defined?

A. 8

B. 21

C. -2

D. 5

616. Nonlinear equations: rational · Hard · SPR

What value of x satisfies $\dfrac{x+23}{x+5}=3$, given that the expression is defined?

617. Nonlinear equations: rational · Hard · SPR

What value of x satisfies $\dfrac{x+30}{x+3}=4$, given that the expression is defined?

618. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x+36}{x+3}=4$, given that the expression is defined?

A. -3

B. 8

C. 44

D. 12

619. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x+24}{x+6}=3$, given that the expression is defined?

A. -6

B. 6

C. 27

D. 3

620. Nonlinear equations: rational · Hard · MC

What value of x satisfies $\dfrac{x+32}{x+2}=4$, given that the expression is defined?

A. 8

B. 12

C. -2

D. 40

621. Nonlinear systems: x-coordinate · Medium · SPR

The graphs of $y=x^2$ and $y=2x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

622. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y=x^2$ and $y=5x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 5

B. $\frac{5}{2}$

C. 25

D. 6

623. Nonlinear systems: x-coordinate · Medium · SPR

The graphs of $y = x^2$ and $y = 14x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

624. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y = x^2$ and $y = 16x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 17

B. 8

C. 256

D. 16

625. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y = x^2$ and $y = 13x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. $\frac{13}{2}$

B. 14

C. 13

D. 169

626. Nonlinear systems: x-coordinate · Medium · SPR

The graphs of $y = x^2$ and $y = 4x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

627. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y = x^2$ and $y = 11x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 11

B. $\frac{11}{2}$

C. 121

D. 12

628. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y = x^2$ and $y = 3x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 9

B. $\frac{3}{2}$

C. 3

D. 4

629. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y = x^2$ and $y = 15x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. $\frac{15}{2}$

B. 15

C. 225

D. 16

630. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y = x^2$ and $y = 7x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 7

B. 8

C. 49

D. $\frac{7}{2}$

631. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y = x^2$ and $y = 10x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 100

B. 10

C. 5

D. 11

632. Nonlinear systems: x-coordinate · Medium · MC

The graphs of $y = x^2$ and $y = 8x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 8

B. 64

C. 4

D. 9

633. Nonlinear systems: y-coordinate · Hard · MC

The graphs of $y = x^2$ and $y = 10x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 0

B. 10

C. 100

D. 20

634. Nonlinear systems: y-coordinate · Hard · MC

The graphs of $y = x^2$ and $y = 8x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 8

B. 16

C. 0

D. 64

635. Nonlinear systems: y-coordinate · Hard · MC

The graphs of $y = x^2$ and $y = 7x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 14

B. 49

C. 0

D. 7

636. Nonlinear systems: y-coordinate · Hard · SPR

The graphs of $y = x^2$ and $y = 4x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

637. Nonlinear systems: y-coordinate · Hard · MC

The graphs of $y = x^2$ and $y = 6x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 12

B. 6

C. 36

D. 0

638. Nonlinear systems: y-coordinate · Hard · MC

The graphs of $y = x^2$ and $y = 2x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 0

B. 2

C. 4

D. 5

639. Nonlinear systems: y-coordinate · Hard · SPR

The graphs of $y = x^2$ and $y = 9x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

640. Nonlinear systems: y-coordinate · Hard · MC

The graphs of $y = x^2$ and $y = 5x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 10

B. 25

C. 0

D. 5

641. Nonlinear systems: y-coordinate · Hard · MC

The graphs of $y = x^2$ and $y = 11x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 0

B. 121

C. 11

D. 22

642. Nonlinear systems: y-coordinate · Hard · MC

The graphs of $y = x^2$ and $y = 12x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 144

B. 12

C. 0

D. 24

643. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-3)(x+3)-3(x-3)$?
A. $(x-3)(x-0)$
B. $(x-3)(x+0)$
C. $(x-3)(x+6)$
D. $(x+3)(x+0)$

644. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-6)(x+3)-2(x-6)$?
A. $(x+6)(x+1)$
B. $(x-6)(x-1)$
C. $(x-6)(x+1)$
D. $(x-6)(x+5)$

645. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-5)(x+7)-2(x-5)$?
A. $(x+5)(x+5)$
B. $(x-5)(x-5)$
C. $(x-5)(x+5)$
D. $(x-5)(x+9)$

646. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-5)(x+4)-2(x-5)$?
A. $(x-5)(x+2)$
B. $(x-5)(x+6)$
C. $(x+5)(x+2)$
D. $(x-5)(x-2)$

647. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-5)(x+5)-2(x-5)$?
A. $(x-5)(x-3)$
B. $(x-5)(x+3)$
C. $(x-5)(x+7)$
D. $(x+5)(x+3)$

648. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-6)(x+5)-5(x-6)$?
A. $(x-6)(x+0)$
B. $(x-6)(x+10)$
C. $(x+6)(x+0)$
D. $(x-6)(x-0)$

649. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-3)(x+4)-4(x-3)$?
A. $(x-3)(x-0)$
B. $(x-3)(x+0)$
C. $(x-3)(x+8)$
D. $(x+3)(x+0)$

650. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-4)(x+4)-2(x-4)$?
A. $(x+4)(x+2)$
B. $(x-4)(x-2)$
C. $(x-4)(x+2)$
D. $(x-4)(x+6)$

651. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-5)(x+7)-5(x-5)$?
A. $(x-5)(x-2)$
B. $(x-5)(x+2)$
C. $(x-5)(x+12)$
D. $(x+5)(x+2)$

652. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-5)(x+7)-3(x-5)$?
A. $(x-5)(x+10)$
B. $(x+5)(x+4)$
C. $(x-5)(x-4)$
D. $(x-5)(x+4)$

653. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-6)(x+2)-4(x-6)$?
A. $(x-6)(x--2)$
B. $(x-6)(x+-2)$
C. $(x-6)(x+6)$
D. $(x+6)(x+-2)$

654. Equivalent expressions: common binomial factor · Hard · MC
Which expression is equivalent to $(x-4)(x+5)-5(x-4)$?
A. $(x-4)(x+0)$
B. $(x-4)(x+10)$
C. $(x+4)(x+0)$
D. $(x-4)(x-0)$

655. Nonlinear equations: absolute value, sum of solutions · Medium · SPR
The equation $|x-12|=6$ has two solutions. What is the sum of those solutions?

656. Nonlinear equations: absolute value, sum of solutions · Medium · SPR
The equation $|x-2|=7$ has two solutions. What is the sum of those solutions?

657. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x-9|=3$ has two solutions. What is the sum of those solutions?
A. 9
B. 18
C. 3
D. 12

658. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x - 6| = 7$ has two solutions. What is the sum of those solutions?
A. 6
B. 13
C. 12
D. 7

659. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x - 7| = 9$ has two solutions. What is the sum of those solutions?
A. 7
B. 9
C. 14
D. 16

660. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x - 9| = 9$ has two solutions. What is the sum of those solutions?
A. 18
B. 19
C. 0
D. 9

661. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x - 10| = 8$ has two solutions. What is the sum of those solutions?
A. 10
B. 8
C. 18
D. 20

662. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x - 7| = 7$ has two solutions. What is the sum of those solutions?
A. 7
B. 14
C. 15
D. 0

663. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x - 6| = 4$ has two solutions. What is the sum of those solutions?
A. 12
B. 6
C. 4
D. 10

664. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x - 8| = 3$ has two solutions. What is the sum of those solutions?
A. 11
B. 16
C. 8
D. 3

665. Nonlinear equations: absolute value, sum of solutions · Medium · SPR
The equation $|x - 6| = 6$ has two solutions. What is the sum of those solutions?

666. Nonlinear equations: absolute value, sum of solutions · Medium · MC
The equation $|x - 4| = 1$ has two solutions. What is the sum of those solutions?
A. 8
B. 1
C. 5
D. 4

667. Nonlinear equations: absolute value, sum of solutions · Medium · SPR
The equation $|x - 2| = 4$ has two solutions. What is the sum of those solutions?

668. Nonlinear equations: absolute value, sum of solutions · Medium · SPR
The equation $|x - 11| = 9$ has two solutions. What is the sum of those solutions?

669. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 8x + c$ has $(x - 2)$ as a factor. What is the value of c?
A. -20
B. 20
C. 2
D. 8

670. Nonlinear equations: factor theorem · Hard · SPR
The polynomial $x^2 + 5x + c$ has $(x - 4)$ as a factor. What is the value of c?

671. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 4x + c$ has $(x - 3)$ as a factor. What is the value of c?
A. 3
B. 21
C. 4
D. -21

672. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 1x + c$ has $(x - 4)$ as a factor. What is the value of c?
A. 4
B. 20
C. 1
D. -20

673. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 7x + c$ has $(x - 2)$ as a factor. What is the value of c?
A. 2
B. 7
C. -18
D. 18

674. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 7x + c$ has $(x - 5)$ as a factor. What is the value of c?
A. 60
B. 7
C. 5
D. -60

675. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 7x + c$ has $(x - 4)$ as a factor. What is the value of c?
A. 44
B. 4
C. 7
D. -44

676. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 2x + c$ has $(x - 4)$ as a factor. What is the value of c?
A. 24
B. 4
C. -24
D. 2

677. Nonlinear equations: factor theorem · Hard · SPR
The polynomial $x^2 + 7x + c$ has $(x - 3)$ as a factor. What is the value of c?

678. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 5x + c$ has $(x - 3)$ as a factor. What is the value of c?
A. 3
B. 5
C. -24
D. 24

679. Nonlinear equations: factor theorem · Hard · SPR
The polynomial $x^2 + 6x + c$ has $(x - 2)$ as a factor. What is the value of c?

680. Nonlinear equations: factor theorem · Hard · MC
The polynomial $x^2 + 6x + c$ has $(x - 3)$ as a factor. What is the value of c?
A. 27
B. 6
C. 3
D. -27

681. Equivalent expressions: perfect square · Easy · MC
Which expression is equivalent to $(x + 3)^2$?
A. $x^2 + 6x + 3$
B. $x^2 + 9x + 6$
C. $x^2 + 6x + 9$
D. $x^2 + 3x + 9$

682. Equivalent expressions: perfect square · Easy · MC
Which expression is equivalent to $(x+7)^2$?
A. $x^2+7x+49$
B. $x^2+14x+7$
C. $x^2+49x+14$
D. $x^2+14x+49$

683. Equivalent expressions: perfect square · Easy · MC
Which expression is equivalent to $(x+9)^2$?
A. $x^2+9x+81$
B. $x^2+18x+9$
C. $x^2+81x+18$
D. $x^2+18x+81$

684. Equivalent expressions: perfect square · Easy · MC
Which expression is equivalent to $(x+6)^2$?
A. $x^2+12x+36$
B. $x^2+6x+36$
C. $x^2+12x+6$
D. $x^2+36x+12$

685. Equivalent expressions: perfect square · Easy · MC
Which expression is equivalent to $(x+14)^2$?
A. $x^2+14x+196$
B. $x^2+28x+14$
C. $x^2+196x+28$
D. $x^2+28x+196$

686. Equivalent expressions: perfect square · Easy · MC
Which expression is equivalent to $(x+13)^2$?
A. $x^2+26x+13$
B. $x^2+169x+26$
C. $x^2+26x+169$
D. $x^2+13x+169$

687. Equivalent expressions: perfect square · Easy · MC
Which expression is equivalent to $(x+12)^2$?
A. $x^2+12x+144$
B. $x^2+24x+12$
C. $x^2+144x+24$
D. $x^2+24x+144$

688. Equivalent expressions: perfect square · Easy · MC
Which expression is equivalent to $(x+11)^2$?
A. $x^2+121x+22$
B. $x^2+22x+121$
C. $x^2+11x+121$
D. $x^2+22x+11$

689. Equivalent expressions: perfect square · Easy · MC

Which expression is equivalent to $(x+5)^2$?

A. $x^2+25x+10$

B. $x^2+10x+25$

C. $x^2+5x+25$

D. $x^2+10x+5$

690. Equivalent expressions: perfect square · Easy · MC

Which expression is equivalent to $(x+10)^2$?

A. $x^2+20x+100$

B. $x^2+10x+100$

C. $x^2+20x+10$

D. $x^2+100x+20$

691. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 2. Which statement must be true?

A. The object rises for 2 seconds and then falls for 2 seconds.

B. The object is at height 0 at 2 seconds.

C. The object's maximum height is 2 meters.

D. The object was launched from a height of 2 meters.

692. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 5. Which statement must be true?

A. The object was launched from a height of 5 meters.

B. The object rises for 5 seconds and then falls for 5 seconds.

C. The object is at height 0 at 5 seconds.

D. The object's maximum height is 5 meters.

693. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 12. Which statement must be true?

A. The object rises for 12 seconds and then falls for 12 seconds.

B. The object is at height 0 at 12 seconds.

C. The object's maximum height is 12 meters.

D. The object was launched from a height of 12 meters.

694. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 3. Which statement must be true?

A. The object's maximum height is 3 meters.

B. The object was launched from a height of 3 meters.

C. The object rises for 3 seconds and then falls for 3 seconds.

D. The object is at height 0 at 3 seconds.

695. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 9. Which statement must be true?

A. The object's maximum height is 9 meters.

B. The object was launched from a height of 9 meters.

C. The object rises for 9 seconds and then falls for 9 seconds.

D. The object is at height 0 at 9 seconds.

696. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 8. Which statement must be true?

A. The object is at height 0 at 8 seconds.

B. The object's maximum height is 8 meters.

C. The object was launched from a height of 8 meters.

D. The object rises for 8 seconds and then falls for 8 seconds.

697. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 11. Which statement must be true?

A. The object is at height 0 at 11 seconds.

B. The object's maximum height is 11 meters.

C. The object was launched from a height of 11 meters.

D. The object rises for 11 seconds and then falls for 11 seconds.

698. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 6. Which statement must be true?

A. The object rises for 6 seconds and then falls for 6 seconds.

B. The object is at height 0 at 6 seconds.

C. The object's maximum height is 6 meters.

D. The object was launched from a height of 6 meters.

699. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 7. Which statement must be true?

A. The object's maximum height is 7 meters.

B. The object was launched from a height of 7 meters.

C. The object rises for 7 seconds and then falls for 7 seconds.

D. The object is at height 0 at 7 seconds.

700. Nonlinear functions: meaning of a zero · Medium · MC

The height of an object above the ground, in meters, is modeled by a function of time in seconds. A positive zero of the function is 4. Which statement must be true?

A. The object was launched from a height of 4 meters.

B. The object rises for 4 seconds and then falls for 4 seconds.

C. The object is at height 0 at 4 seconds.

D. The object's maximum height is 4 meters.

Problem-Solving and Data Analysis

701. Ratios: part to whole · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 3. There are 6 red marbles. How many marbles are in the bag?

A. 18

B. 9

C. 12

D. 6

702. Ratios: part to whole · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 2. There are 12 red marbles. How many marbles are in the bag?

A. 20

B. 14

C. 6

D. 24

703. Ratios: part to whole · Easy · SPR

The ratio of red marbles to blue marbles in a bag is 2 : 3. There are 10 red marbles. How many marbles are in the bag?

704. Ratios: part to whole · Easy · MC

The ratio of red marbles to blue marbles in a bag is 2 : 2. There are 12 red marbles. How many marbles are in the bag?

A. 25

B. 14

C. 4

D. 24

705. Ratios: part to whole · Easy · MC

The ratio of red marbles to blue marbles in a bag is 2 : 6. There are 4 red marbles. How many marbles are in the bag?

A. 16

B. 24

C. 12

D. 10

706. Ratios: part to whole · Easy · SPR

The ratio of red marbles to blue marbles in a bag is 2 : 5. There are 4 red marbles. How many marbles are in the bag?

707. Ratios: part to whole · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 6. There are 18 red marbles. How many marbles are in the bag?

A. 18

B. 108

C. 54

D. 24

708. Ratios: part to whole · Easy · MC

The ratio of red marbles to blue marbles in a bag is 2 : 5. There are 6 red marbles. How many marbles are in the bag?

A. 30

B. 11

C. 10

D. 21

709. Ratios: part to whole · Easy · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 6. There are 8 red marbles. How many marbles are in the bag?

710. Ratios: part to whole · Easy · SPR

The ratio of red marbles to blue marbles in a bag is 5 : 5. There are 15 red marbles. How many marbles are in the bag?

711. Rates: distance and time · Easy · MC

A train travels 296 miles in 4 hours at a constant speed. What is the speed in miles per hour?

A. 4

B. 78

C. 296

D. 74

712. Rates: distance and time · Easy · MC

A train travels 88 miles in 2 hours at a constant speed. What is the speed in miles per hour?

A. 88

B. 44

C. 46

D. 2

713. Rates: distance and time · Easy · SPR

A train travels 295 miles in 5 hours at a constant speed. What is the speed in miles per hour?

714. Rates: distance and time · Easy · MC

A train travels 300 miles in 5 hours at a constant speed. What is the speed in miles per hour?

A. 300

B. 5

C. 65

D. 60

715. Rates: distance and time · Easy · SPR

A train travels 86 miles in 2 hours at a constant speed. What is the speed in miles per hour?

716. Rates: distance and time · Easy · MC

A train travels 270 miles in 5 hours at a constant speed. What is the speed in miles per hour?

A. 5

B. 54

C. 59

D. 270

717. Rates: distance and time · Easy · MC

A train travels 120 miles in 3 hours at a constant speed. What is the speed in miles per hour?

A. 40

B. 120

C. 43

D. 3

718. Rates: distance and time · Easy · MC

A train travels 276 miles in 4 hours at a constant speed. What is the speed in miles per hour?

A. 69

B. 4

C. 73

D. 276

719. Rates: scale · Medium · SPR

On a map, 1 centimeter represents 3 kilometers. A road measures 12 centimeters on the map. What is the actual length of the road, in kilometers?

720. Rates: scale · Medium · MC

On a map, 1 centimeter represents 2 kilometers. A road measures 9 centimeters on the map. What is the actual length of the road, in kilometers?

A. 11

B. 18

C. 9

D. 2

721. Rates: scale · Medium · SPR

On a map, 1 centimeter represents 3 kilometers. A road measures 8 centimeters on the map. What is the actual length of the road, in kilometers?

722. Rates: scale · Medium · MC

On a map, 1 centimeter represents 8 kilometers. A road measures 10 centimeters on the map. What is the actual length of the road, in kilometers?

A. 8

B. 18

C. 80

D. 10

723. Rates: scale · Medium · MC

On a map, 1 centimeter represents 4 kilometers. A road measures 11 centimeters on the map. What is the actual length of the road, in kilometers?

A. 15

B. 4

C. 44

D. 11

724. Rates: scale · Medium · SPR

On a map, 1 centimeter represents 5 kilometers. A road measures 3 centimeters on the map. What is the actual length of the road, in kilometers?

725. Rates: scale · Medium · MC

On a map, 1 centimeter represents 8 kilometers. A road measures 4 centimeters on the map. What is the actual length of the road, in kilometers?

A. 12

B. 32

C. 4

D. 8

726. Rates: scale · Medium · SPR

On a map, 1 centimeter represents 4 kilometers. A road measures 7 centimeters on the map. What is the actual length of the road, in kilometers?

727. Proportions: scale a recipe · Medium · MC

A recipe that serves 4 people uses 2 cups of rice. How many cups of rice are needed to serve 8 people at the same rate?

A. 6

B. 8

C. 2

D. 4

728. Proportions: scale a recipe · Medium · MC

A recipe that serves 8 people uses 4 cups of rice. How many cups of rice are needed to serve 24 people at the same rate?

A. 12

B. 4

C. 7

D. 24

729. Proportions: scale a recipe · Medium · MC

A recipe that serves 4 people uses 4 cups of rice. How many cups of rice are needed to serve 8 people at the same rate?

A. 4

B. 12

C. 8

D. 6

730. Proportions: scale a recipe · Medium · MC

A recipe that serves 6 people uses 3 cups of rice. How many cups of rice are needed to serve 12 people at the same rate?

A. 12

B. 5

C. 6

D. 3

731. Proportions: scale a recipe · Medium · MC

A recipe that serves 6 people uses 3 cups of rice. How many cups of rice are needed to serve 18 people at the same rate?

A. 18

B. 3

C. 6

D. 9

732. Proportions: scale a recipe · Medium · MC

A recipe that serves 6 people uses 2 cups of rice. How many cups of rice are needed to serve 18 people at the same rate?

A. 2

B. 6

C. 5

D. 18

733. Proportions: scale a recipe · Medium · SPR

A recipe that serves 8 people uses 2 cups of rice. How many cups of rice are needed to serve 16 people at the same rate?

734. Proportions: scale a recipe · Medium · SPR

A recipe that serves 4 people uses 6 cups of rice. How many cups of rice are needed to serve 12 people at the same rate?

735. Percentages: discount · Easy · SPR

A jacket costs \$80. During a sale the price is reduced by 10%. What is the sale price, in dollars?

736. Percentages: discount · Easy · MC

A jacket costs \$60. During a sale the price is reduced by 25%. What is the sale price, in dollars?

A. 45

B. 35

C. 85

D. 15

737. Percentages: discount · Easy · SPR

A jacket costs \$100. During a sale the price is reduced by 30%. What is the sale price, in dollars?

738. Percentages: discount · Easy · MC

A jacket costs \$100. During a sale the price is reduced by 15%. What is the sale price, in dollars?

A. 115

B. 85

C. 86

D. 15

739. Percentages: discount · Easy · MC

A jacket costs \$100. During a sale the price is reduced by 10%. What is the sale price, in dollars?

A. 10

B. 91

C. 110

D. 90

740. Percentages: discount · Easy · MC

A jacket costs \$200. During a sale the price is reduced by 25%. What is the sale price, in dollars?

A. 225

B. 50

C. 175

D. 150

741. Percentages: discount · Easy · MC

A jacket costs $40. During a sale the price is reduced by 10%. What is the sale price, in dollars?

A. 50

B. 36

C. 30

D. 4

742. Percentages: discount · Easy · MC

A jacket costs $60. During a sale the price is reduced by 15%. What is the sale price, in dollars?

A. 75

B. 45

C. 9

D. 51

743. Percentages: discount · Easy · MC

A jacket costs $80. During a sale the price is reduced by 20%. What is the sale price, in dollars?

A. 100

B. 16

C. 64

D. 60

744. Percentages: discount · Easy · SPR

A jacket costs $40. During a sale the price is reduced by 30%. What is the sale price, in dollars?

745. Percentages: increase · Medium · MC

A fee of $80 increases by 25%. What is the new fee, in dollars?

A. 20

B. 100

C. 55

D. 105

746. Percentages: increase · Medium · SPR

A fee of $120 increases by 25%. What is the new fee, in dollars?

747. Percentages: increase · Medium · MC

A fee of $100 increases by 15%. What is the new fee, in dollars?

A. 15

B. 115

C. 116

D. 85

748. Percentages: increase · Medium · MC

A fee of $80 increases by 20%. What is the new fee, in dollars?

A. 100

B. 60

C. 96

D. 16

749. Percentages: increase · Medium · MC

A fee of \$60 increases by 15%. What is the new fee, in dollars?

A. 9

B. 69

C. 45

D. 75

750. Percentages: increase · Medium · MC

A fee of \$60 increases by 20%. What is the new fee, in dollars?

A. 40

B. 80

C. 72

D. 12

751. Percentages: increase · Medium · SPR

A fee of \$120 increases by 20%. What is the new fee, in dollars?

752. Percentages: increase · Medium · MC

A fee of \$40 increases by 25%. What is the new fee, in dollars?

A. 65

B. 15

C. 10

D. 50

753. Percentages: discount then tax · Hard · SPR

A bike is listed at \$50. It is discounted by 20%, and then a 20% sales tax is added to the discounted price. What is the final price, in dollars?

754. Percentages: discount then tax · Hard · MC

A bike is listed at \$60. It is discounted by 25%, and then a 20% sales tax is added to the discounted price. What is the final price, in dollars?

A. 54

B. 35

C. 45

D. 65

755. Percentages: discount then tax · Hard · MC

A bike is listed at \$200. It is discounted by 50%, and then a 20% sales tax is added to the discounted price. What is the final price, in dollars?

A. 120

B. 150

C. 100

D. 200

756. Percentages: discount then tax · Hard · MC

A bike is listed at \$200. It is discounted by 40%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 120

B. 132

C. 160

D. 130

757. Percentages: discount then tax · Hard · SPR

A bike is listed at \$200. It is discounted by 25%, and then a 8% sales tax is added to the discounted price. What is the final price, in dollars?

758. Percentages: discount then tax · Hard · SPR

A bike is listed at \$120. It is discounted by 25%, and then a 20% sales tax is added to the discounted price. What is the final price, in dollars?

759. Percentages: discount then tax · Hard · MC

A bike is listed at \$200. It is discounted by 50%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 100

B. 110

C. 150

D. 200

760. Percentages: discount then tax · Hard · MC

A bike is listed at \$120. It is discounted by 50%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 70

B. 120

C. 60

D. 66

761. One-variable data: mean · Easy · MC

The mean of the numbers 8, 10, 12, 9, and x is 8. What is the value of x?

A. 3

B. 8

C. 9

D. 1

762. One-variable data: mean · Easy · MC

The mean of the numbers 6, 4, 15, 9, and x is 13. What is the value of x?

A. 31

B. 13

C. 8

D. 33

763. One-variable data: mean · Easy · MC

The mean of the numbers 5, 16, 14, 9, and x is 14. What is the value of x?

A. 14

B. 26

C. 28

D. 11

764. One-variable data: mean · Easy · MC

The mean of the numbers 5, 11, 6, 4, and x is 11. What is the value of x?

A. 6

B. 31

C. 11

D. 29

765. One-variable data: mean · Easy · SPR
The mean of the numbers 9, 10, 15, 15, and x is 12. What is the value of x?

766. One-variable data: mean · Easy · MC
The mean of the numbers 15, 6, 4, 17, and x is 15. What is the value of x?
A. 35
B. 10
C. 15
D. 33

767. One-variable data: mean · Easy · MC
The mean of the numbers 9, 12, 17, 12, and x is 13. What is the value of x?
A. 17
B. 15
C. 12
D. 13

768. One-variable data: mean · Easy · MC
The mean of the numbers 12, 14, 17, 12, and x is 15. What is the value of x?
A. 13
B. 20
C. 22
D. 15

769. One-variable data: median · Easy · MC
What is the median of the data set 11, 16, 19, 19, 22?
A. 19
B. 16
C. 22
D. 11

770. One-variable data: median · Easy · MC
What is the median of the data set 9, 21, 21, 24, 26?
A. 26
B. 9
C. 21
D. 20

771. One-variable data: median · Easy · SPR
What is the median of the data set 2, 10, 14, 19, 29?

772. One-variable data: median · Easy · SPR
What is the median of the data set 12, 16, 16, 25, 27?

773. One-variable data: median · Easy · SPR
What is the median of the data set 2, 8, 8, 21, 29?

774. One-variable data: median · Easy · MC
What is the median of the data set 6, 15, 16, 24, 27?
A. 6
B. 27
C. 16
D. 15

775. One-variable data: median · Easy · MC
What is the median of the data set 3, 16, 17, 20, 21?
A. 21
B. 3
C. 16
D. 17

776. One-variable data: median · Easy · SPR
What is the median of the data set 13, 21, 21, 22, 24?

777. One-variable data: range · Medium · MC
What is the range of the data set 7, 23, 26, 28, 38?
A. 26
B. 38
C. 7
D. 31

778. One-variable data: range · Medium · MC
What is the range of the data set 5, 8, 12, 16, 31?
A. 26
B. 31
C. 12
D. 5

779. One-variable data: range · Medium · MC
What is the range of the data set 5, 7, 19, 21, 24?
A. 5
B. 19
C. 20
D. 24

780. One-variable data: range · Medium · MC
What is the range of the data set 14, 16, 21, 31, 38?
A. 24
B. 38
C. 14
D. 21

781. One-variable data: range · Medium · MC

What is the range of the data set 2, 11, 30, 37, 39?

A. 30

B. 2

C. 39

D. 37

782. One-variable data: range · Medium · MC

What is the range of the data set 13, 16, 19, 31, 37?

A. 19

B. 37

C. 13

D. 24

783. Probability: one draw · Easy · MC

A bag has 4 red, 5 blue, and 2 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{4}{5}$

B. $\frac{5}{11}$

C. $\frac{1}{3}$

D. $\frac{4}{11}$

784. Probability: one draw · Easy · MC

A bag has 5 red, 6 blue, and 3 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{5}{6}$

B. $\frac{5}{14}$

C. $\frac{3}{7}$

D. $\frac{1}{3}$

785. Probability: one draw · Easy · SPR

A bag has 4 red, 4 blue, and 6 green counters. One counter is drawn at random. What is the probability that it is red?

786. Probability: one draw · Easy · MC

A bag has 7 red, 7 blue, and 2 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{7}{16}$

B. 1

C. $\frac{23}{16}$

D. $\frac{1}{3}$

787. Probability: one draw · Easy · SPR

A bag has 9 red, 3 blue, and 3 green counters. One counter is drawn at random. What is the probability that it is red?

788. Probability: one draw · Easy · MC

A bag has 6 red, 2 blue, and 6 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{7}$

B. $\frac{3}{7}$

C. $\frac{1}{3}$

D. 3

789. Probability: one draw · Easy · MC

A bag has 3 red, 4 blue, and 2 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. $\frac{1}{3}$

D. $\frac{4}{9}$

790. Probability: one draw · Easy · MC

A bag has 4 red, 7 blue, and 3 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{2}{7}$

D. $\frac{4}{7}$

791. Probability: conditional · Hard · MC

A bag has 7 red, 2 blue, and 5 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{5}{14}$

B. $\frac{1}{2}$

C. $\frac{7}{9}$

D. $\frac{2}{9}$

792. Probability: conditional · Hard · MC

A bag has 7 red, 5 blue, and 3 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{5}{12}$

B. $\frac{7}{12}$

C. $\frac{1}{5}$

D. $\frac{7}{15}$

793. Probability: conditional · Hard · MC

A bag has 7 red, 2 blue, and 3 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{2}{9}$

B. $\frac{1}{4}$

C. $\frac{7}{12}$

D. $\frac{7}{9}$

794. Probability: conditional · Hard · MC

A bag has 8 red, 4 blue, and 5 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{2}{3}$

B. $\frac{5}{17}$

C. $\frac{1}{3}$

D. $\frac{8}{17}$

795. Probability: conditional · Hard · MC

A bag has 3 red, 6 blue, and 5 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{3}{14}$

D. $\frac{5}{14}$

796. Probability: conditional · Hard · SPR

A bag has 5 red, 2 blue, and 3 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

797. Probability: complement · Medium · MC

The probability that a randomly selected marble is red is 2/10. What is the probability that it is not red?

A. $\frac{1}{4}$

B. $\frac{4}{5}$

C. $\frac{1}{5}$

D. $\frac{1}{10}$

798. Probability: complement · Medium · MC

The probability that a randomly selected marble is red is 1/5. What is the probability that it is not red?

A. $\frac{4}{5}$

B. $\frac{9}{5}$

C. $\frac{1}{5}$

D. $\frac{1}{4}$

799. Probability: complement · Medium · MC

The probability that a randomly selected marble is red is 4/6. What is the probability that it is not red?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. 2

D. $\frac{2}{3}$

800. Probability: complement · Medium · SPR

The probability that a randomly selected marble is red is 3/6. What is the probability that it is not red?

801. Probability: complement · Medium · MC

The probability that a randomly selected marble is red is 2/6. What is the probability that it is not red?

A. $\frac{1}{6}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

802. Probability: complement · Medium · MC

The probability that a randomly selected marble is red is 2/5. What is the probability that it is not red?

A. $\frac{2}{5}$

B. $\frac{2}{3}$

C. $\frac{1}{5}$

D. $\frac{3}{5}$

803. Probability: two-way table · Medium · SPR

Among juniors, 15 joined and 9 did not. Among seniors, 5 joined and 9 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

804. Probability: two-way table · Medium · MC

Among juniors, 18 joined and 8 did not. Among seniors, 13 joined and 9 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{3}{8}$

B. $\frac{18}{31}$

C. $\frac{13}{31}$

D. $\frac{9}{13}$

805. Probability: two-way table · Medium · SPR

Among juniors, 13 joined and 16 did not. Among seniors, 5 joined and 13 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

806. Probability: two-way table · Medium · MC

Among juniors, 17 joined and 15 did not. Among seniors, 11 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{17}{28}$

B. $\frac{17}{32}$

C. $\frac{1}{3}$

D. $\frac{11}{28}$

807. Probability: two-way table · Medium · MC

Among juniors, 8 joined and 7 did not. Among seniors, 5 joined and 12 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{1}{4}$

B. $\frac{5}{13}$

C. $\frac{8}{15}$

D. $\frac{8}{13}$

808. Probability: two-way table · Medium · MC

Among juniors, 8 joined and 16 did not. Among seniors, 12 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{2}{11}$

B. $\frac{3}{5}$

C. $\frac{2}{5}$

D. $\frac{1}{3}$

809. Probability: two-way table · Medium · MC

Among juniors, 8 joined and 14 did not. Among seniors, 12 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{4}{21}$

B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. $\frac{4}{11}$

810. Probability: two-way table · Medium · MC

Among juniors, 17 joined and 10 did not. Among seniors, 12 joined and 12 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{17}{27}$

B. $\frac{12}{29}$

C. $\frac{1}{3}$

D. $\frac{17}{29}$

811. Two-variable data: predict from a model · Medium · MC

A line of best fit for a data set is $y = 2x + 9$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 6$?

A. 9

B. 21

C. 12

D. 3

812. Two-variable data: predict from a model · Medium · MC

A line of best fit for a data set is $y = -3x + 11$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

A. -32

B. 11

C. -10

D. -21

813. Two-variable data: predict from a model · Medium · SPR

A line of best fit for a data set is $y = 4x + 11$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 9$?

814. Two-variable data: predict from a model · Medium · MC

A line of best fit for a data set is $y = 2x + 3$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 9$?

A. 3

B. 15

C. 18

D. 21

815. Two-variable data: predict from a model · Medium · MC

A line of best fit for a data set is $y = -3x + 9$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

A. -30

B. -21

C. 9

D. -12

816. Two-variable data: predict from a model · Medium · MC

A line of best fit for a data set is $y = 3x + 1$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

A. 21

B. 20

C. 1

D. 22

817. Two-variable data: predict from a model · Medium · MC

A line of best fit for a data set is $y = 5x + 3$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 9$?

A. 42

B. 48

C. 3

D. 45

818. Two-variable data: predict from a model · Medium · MC

A line of best fit for a data set is $y = 5x + 6$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 8$?

A. 6

B. 46

C. 40

D. 34

819. Two-variable data: meaning of the slope · Medium · MC

A line of best fit is $y = 3x + 26$, where x is hours studied and y is the predicted score. What does 3 represent?

A. The predicted score decreases by 3 points for each additional hour studied

B. A student who studies 0 hours is predicted to score 3

C. The highest score in the data set is 3

D. The predicted score increases by 3 points for each additional hour studied

820. Two-variable data: meaning of the slope · Medium · MC

A line of best fit is $y = 5x + 25$, where x is hours studied and y is the predicted score. What does 5 represent?

A. The highest score in the data set is 5

B. The predicted score increases by 5 points for each additional hour studied

C. The predicted score decreases by 5 points for each additional hour studied

D. A student who studies 0 hours is predicted to score 5

821. Two-variable data: meaning of the slope · Medium · MC

A line of best fit is $y = 4x + 24$, where x is hours studied and y is the predicted score. What does 4 represent?

A. The predicted score increases by 4 points for each additional hour studied

B. The predicted score decreases by 4 points for each additional hour studied

C. A student who studies 0 hours is predicted to score 4

D. The highest score in the data set is 4

822. Two-variable data: meaning of the slope · Medium · MC

A line of best fit is $y = 3x + 13$, where x is hours studied and y is the predicted score. What does 3 represent?

A. The predicted score increases by 3 points for each additional hour studied

B. The predicted score decreases by 3 points for each additional hour studied

C. A student who studies 0 hours is predicted to score 3

D. The highest score in the data set is 3

823. Two-variable data: meaning of the slope · Medium · MC

A line of best fit is $y = 2x + 31$, where x is hours studied and y is the predicted score. What does 2 represent?

A. The highest score in the data set is 2

B. The predicted score increases by 2 points for each additional hour studied

C. The predicted score decreases by 2 points for each additional hour studied

D. A student who studies 0 hours is predicted to score 2

824. Two-variable data: meaning of the slope · Medium · MC

A line of best fit is $y = 5x + 32$, where x is hours studied and y is the predicted score. What does 5 represent?

A. The predicted score increases by 5 points for each additional hour studied

B. The predicted score decreases by 5 points for each additional hour studied

C. A student who studies 0 hours is predicted to score 5

D. The highest score in the data set is 5

825. Inference: margin of error · Medium · MC

In a random sample of 900 voters, 61% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

A. Exactly 61% of all voters support the proposal.

B. The margin of error would stay 5 percentage points if the sample size were much smaller.

C. The result proves that the proposal will pass.

D. It is plausible that between 56% and 66% of all voters support the proposal.

826. Inference: margin of error · Medium · MC

In a random sample of 900 voters, 57% supported a proposal. The margin of error for this estimate is 4 percentage points. Which conclusion is best supported?

A. It is plausible that between 53% and 61% of all voters support the proposal.

B. Exactly 57% of all voters support the proposal.

C. The margin of error would stay 4 percentage points if the sample size were much smaller.

D. The result proves that the proposal will pass.

827. Inference: margin of error · Medium · MC

In a random sample of 400 voters, 52% supported a proposal. The margin of error for this estimate is 4 percentage points. Which conclusion is best supported?

A. The result proves that the proposal will pass.

B. It is plausible that between 48% and 56% of all voters support the proposal.

C. Exactly 52% of all voters support the proposal.

D. The margin of error would stay 4 percentage points if the sample size were much smaller.

828. Inference: margin of error · Medium · MC

In a random sample of 800 voters, 61% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

A. The margin of error would stay 3 percentage points if the sample size were much smaller.

B. The result proves that the proposal will pass.

C. It is plausible that between 58% and 64% of all voters support the proposal.

D. Exactly 61% of all voters support the proposal.

829. Inference: margin of error · Medium · MC

In a random sample of 1200 voters, 57% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

A. Exactly 57% of all voters support the proposal.

B. The margin of error would stay 5 percentage points if the sample size were much smaller.

C. The result proves that the proposal will pass.

D. It is plausible that between 52% and 62% of all voters support the proposal.

830. Inference: margin of error · Medium · MC

In a random sample of 1200 voters, 63% supported a proposal. The margin of error for this estimate is 4 percentage points. Which conclusion is best supported?

A. Exactly 63% of all voters support the proposal.

B. The margin of error would stay 4 percentage points if the sample size were much smaller.

C. The result proves that the proposal will pass.

D. It is plausible that between 59% and 67% of all voters support the proposal.

831. Inference: margin of error · Medium · MC

In a random sample of 800 voters, 63% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

A. The margin of error would stay 3 percentage points if the sample size were much smaller.

B. The result proves that the proposal will pass.

C. It is plausible that between 60% and 66% of all voters support the proposal.

D. Exactly 63% of all voters support the proposal.

832. Inference: margin of error · Medium · MC

In a random sample of 900 voters, 52% supported a proposal. The margin of error for this estimate is 4 percentage points. Which conclusion is best supported?

A. The margin of error would stay 4 percentage points if the sample size were much smaller.

B. The result proves that the proposal will pass.

C. It is plausible that between 48% and 56% of all voters support the proposal.

D. Exactly 52% of all voters support the proposal.

833. Statistical claims: experiment or observation · Hard · MC

To estimate the proportion of all students at a college who live on campus, which method is most appropriate?

A. Survey students at a different college.

B. Take a random sample of students from the college and record whether each lives on campus.

C. Ask only the students in one dormitory.

D. Assign a random half of the college to move on campus.

834. Statistical claims: experiment or observation · Hard · MC

Researchers randomly select classrooms, then randomly assign those classrooms to a new reading program or the current program. The random assignment is useful mainly because it helps

A. make the margin of error equal to zero.

B. support a conclusion about the effect of the program.

C. remove the need to record any scores.

D. guarantee that every student in the country is represented.

835. Statistical claims: experiment or observation · Hard · MC

A school claims that students who join the chess club have higher math grades, so the club causes the higher grades. Why is that claim not well supported?

A. Observational results are never allowed to include numbers.

B. Students who choose the club may already differ from students who do not.

C. The school did not report the margin of error of a poll.

D. A mean cannot be computed from letter grades.

836. Statistical claims: experiment or observation · Hard · MC

A coach wants to test whether a new warm-up causes faster sprint times. Which design best supports a cause-and-effect conclusion?

A. Use the new warm-up for the whole team and compare this season with last season.

B. Randomly assign athletes to the new warm-up or the usual warm-up, then compare sprint times.

C. Let the fastest athletes choose the new warm-up and compare them with the others.

D. Survey athletes about which warm-up they like.

837. Statistical claims: experiment or observation · Hard · MC

A researcher wants to know whether a new homework format causes higher quiz scores at one school. Which design best supports a cause-and-effect conclusion?

A. Compare this year's quiz scores with last year's after the school changes the format for everyone.

B. Randomly assign students at the school to the new format or the current format, then compare quiz scores.

C. Ask volunteers to try the new format and compare them with students who kept the old format.

D. Survey a random sample of students about which format they prefer.

838. Statistical claims: experiment or observation · Hard · MC

A garden club says members have healthier plants than nonmembers, so membership causes healthier plants. Why is that claim not well supported?

A. People who choose to join may already garden differently from people who do not.

B. A percent cannot be computed from a plant.

C. The club did not name the margin of error.

D. An experiment is never allowed to use a control group.

839. Inference: what shrinks a margin of error · Easy · MC

A school surveys a random sample of students about a new schedule. Which change would be expected to reduce the margin of error?

A. Report the result in words instead of numbers.

B. Survey a larger random sample.

C. Ask only the students who already like the schedule.

D. Change the font on the questionnaire.

840. Inference: what shrinks a margin of error · Easy · MC

A poll uses a random sample of city residents. Which change would be expected to reduce the margin of error?

A. Replace the random sample with volunteers.

B. $Report the result as a fraction instead of a percent.$

C. Increase the sample size.

D. Ask the question in a different order only.

841. Inference: what shrinks a margin of error · Easy · MC

A researcher estimates a population percent from a random sample. Which change would be expected to reduce the margin of error?

A. Use a smaller random sample.

B. Switch from a random sample to a convenience sample.

C. Leave the sample size the same and drop the percent sign.

D. Use a larger random sample from the same population.

842. Inference: what shrinks a margin of error · Easy · MC

A city estimates the percent of households with a garden from a random sample. Which change would be expected to reduce the margin of error?

A. Ask the question twice to the same households and average the words.

B. Round the percent to the nearest ten.

C. Increase the number of households in the random sample.

D. Replace the random sample with people who call in.

843. Units: per minute to per hour · Easy · MC

A printer produces 7 pages each minute at a constant rate. How many pages does it produce in one hour?

A. 60

B. 420

C. 67

D. 7

844. Units: per minute to per hour · Easy · MC
A printer produces 5 pages each minute at a constant rate. How many pages does it produce in one hour?
A. 60
B. 65
C. 300
D. 5

845. Units: per minute to per hour · Easy · MC
A printer produces 13 pages each minute at a constant rate. How many pages does it produce in one hour?
A. 60
B. 780
C. 13
D. 73

846. Units: per minute to per hour · Easy · MC
A printer produces 4 pages each minute at a constant rate. How many pages does it produce in one hour?
A. 60
B. 240
C. 4
D. 64

847. Units: per minute to per hour · Easy · MC
A printer produces 3 pages each minute at a constant rate. How many pages does it produce in one hour?
A. 63
B. 180
C. 3
D. 60

848. Units: per minute to per hour · Easy · MC
A printer produces 2 pages each minute at a constant rate. How many pages does it produce in one hour?
A. 62
B. 120
C. 60
D. 2

849. Units: per minute to per hour · Easy · MC
A printer produces 6 pages each minute at a constant rate. How many pages does it produce in one hour?
A. 66
B. 60
C. 360
D. 6

850. Units: per minute to per hour · Easy · MC
A printer produces 11 pages each minute at a constant rate. How many pages does it produce in one hour?
A. 11
B. 660
C. 71
D. 60

Geometry and Trigonometry

851. Triangles: third angle · Easy · SPR
Two angles of a triangle measure 53° and 45°. What is the measure, in degrees, of the third angle?

852. Triangles: third angle · Easy · MC
Two angles of a triangle measure 65° and 58°. What is the measure, in degrees, of the third angle?
A. 57
B. 123
C. 237
D. 90

853. Triangles: third angle · Easy · MC
Two angles of a triangle measure 79° and 52°. What is the measure, in degrees, of the third angle?
A. 49
B. 90
C. 229
D. 131

854. Triangles: third angle · Easy · MC
Two angles of a triangle measure 57° and 46°. What is the measure, in degrees, of the third angle?
A. 103
B. 90
C. 257
D. 77

855. Triangles: third angle · Easy · MC
Two angles of a triangle measure 75° and 32°. What is the measure, in degrees, of the third angle?
A. 73
B. 107
C. 253
D. 90

856. Triangles: third angle · Easy · SPR
Two angles of a triangle measure 59° and 48°. What is the measure, in degrees, of the third angle?

857. Triangles: third angle · Easy · MC
Two angles of a triangle measure 56° and 65°. What is the measure, in degrees, of the third angle?
A. 239
B. 121
C. 90
D. 59

858. Triangles: third angle · Easy · MC
Two angles of a triangle measure 80° and 38°. What is the measure, in degrees, of the third angle?
A. 118
B. 242
C. 90
D. 62

859. Triangles: third angle · Easy · MC

Two angles of a triangle measure 31° and 52°. What is the measure, in degrees, of the third angle?

A. 277

B. 83

C. 90

D. 97

860. Triangles: third angle · Easy · MC

Two angles of a triangle measure 76° and 64°. What is the measure, in degrees, of the third angle?

A. 40

B. 140

C. 220

D. 90

861. Parallel lines: consecutive interior angle · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 58°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 90

B. 58

C. 122

D. 302

862. Parallel lines: consecutive interior angle · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 44°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 136

B. 44

C. 316

D. 90

863. Parallel lines: consecutive interior angle · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 134°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 226

B. 90

C. 46

D. 134

864. Parallel lines: consecutive interior angle · Medium · SPR

Two parallel lines are cut by a transversal. One interior angle measures 63°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

865. Parallel lines: consecutive interior angle · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 79°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 90

B. 281

C. 79

D. 101

866. Parallel lines: consecutive interior angle · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 141°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 141

B. 39

C. 219

D. 90

867. Parallel lines: consecutive interior angle · Medium · SPR

Two parallel lines are cut by a transversal. One interior angle measures 117°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

868. Parallel lines: consecutive interior angle · Medium · SPR

Two parallel lines are cut by a transversal. One interior angle measures 111°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

869. Parallel lines: alternate interior angle · Medium · SPR

Two parallel lines are cut by a transversal. One angle measures 91°. What is the measure, in degrees, of the alternate interior angle?

870. Parallel lines: alternate interior angle · Medium · MC

Two parallel lines are cut by a transversal. One angle measures 77°. What is the measure, in degrees, of the alternate interior angle?

A. 283

B. 103

C. 77

D. 90

871. Parallel lines: alternate interior angle · Medium · SPR

Two parallel lines are cut by a transversal. One angle measures 141°. What is the measure, in degrees, of the alternate interior angle?

872. Parallel lines: alternate interior angle · Medium · SPR

Two parallel lines are cut by a transversal. One angle measures 125°. What is the measure, in degrees, of the alternate interior angle?

873. Parallel lines: alternate interior angle · Medium · SPR

Two parallel lines are cut by a transversal. One angle measures 130°. What is the measure, in degrees, of the alternate interior angle?

874. Parallel lines: alternate interior angle · Medium · SPR

Two parallel lines are cut by a transversal. One angle measures 69°. What is the measure, in degrees, of the alternate interior angle?

875. Parallel lines: corresponding angle · Easy · MC

Two parallel lines are cut by a transversal. One angle measures 118°. What is the measure, in degrees, of the corresponding angle?

A. 90

B. 118

C. 62

D. 242

876. Parallel lines: corresponding angle · Easy · SPR

Two parallel lines are cut by a transversal. One angle measures 107°. What is the measure, in degrees, of the corresponding angle?

877. Parallel lines: corresponding angle · Easy · MC

Two parallel lines are cut by a transversal. One angle measures 136°. What is the measure, in degrees, of the corresponding angle?

A. 90

B. 224

C. 44

D. 136

878. Parallel lines: corresponding angle · Easy · MC

Two parallel lines are cut by a transversal. One angle measures 113°. What is the measure, in degrees, of the corresponding angle?

A. 247

B. 113

C. 90

D. 67

879. Angles: vertical angles · Easy · SPR

Two lines intersect and form a 68° angle. What is the measure, in degrees, of the vertical angle?

880. Angles: vertical angles · Easy · MC

Two lines intersect and form a 75° angle. What is the measure, in degrees, of the vertical angle?

A. 105

B. 285

C. 90

D. 75

881. Angles: vertical angles · Easy · MC

Two lines intersect and form a 27° angle. What is the measure, in degrees, of the vertical angle?

A. 333

B. 90

C. 27

D. 153

882. Angles: vertical angles · Easy · MC

Two lines intersect and form a 79° angle. What is the measure, in degrees, of the vertical angle?

A. 281

B. 79

C. 90

D. 101

883. Right triangles: missing leg · Medium · MC

A right triangle has a hypotenuse of length 41 and one leg of length 9. What is the length of the other leg?

A. 32

B. 40

C. 50

D. 41

884. Right triangles: missing leg · Medium · MC
A right triangle has a hypotenuse of length 29 and one leg of length 20. What is the length of the other leg?
A. 29
B. 49
C. 9
D. 21

885. Right triangles: missing leg · Medium · MC
A right triangle has a hypotenuse of length 17 and one leg of length 8. What is the length of the other leg?
A. 17
B. 15
C. 9
D. 25

886. Right triangles: missing leg · Medium · MC
A right triangle has a hypotenuse of length 5 and one leg of length 3. What is the length of the other leg?
A. 4
B. 2
C. 5
D. 8

887. Right triangles: missing leg · Medium · MC
A right triangle has a hypotenuse of length 10 and one leg of length 6. What is the length of the other leg?
A. 10
B. 16
C. 8
D. 4

888. Right triangles: missing leg · Medium · SPR
A right triangle has a hypotenuse of length 37 and one leg of length 12. What is the length of the other leg?

889. Right triangles: missing leg · Medium · SPR
A right triangle has a hypotenuse of length 25 and one leg of length 7. What is the length of the other leg?

890. Right triangles: missing leg · Medium · MC
A right triangle has a hypotenuse of length 13 and one leg of length 5. What is the length of the other leg?
A. 12
B. 18
C. 8
D. 13

891. Right triangles: hypotenuse · Hard · MC
A right triangle has legs of length 7 and 24. What is the length of the hypotenuse?
A. 168
B. 25
C. 31
D. 17

892. Right triangles: hypotenuse · Hard · MC

A right triangle has legs of length 8 and 15. What is the length of the hypotenuse?

A. 7

B. 17

C. 23

D. 120

893. Right triangles: hypotenuse · Hard · SPR

A right triangle has legs of length 5 and 12. What is the length of the hypotenuse?

894. Right triangles: hypotenuse · Hard · MC

A right triangle has legs of length 9 and 12. What is the length of the hypotenuse?

A. 15

B. 3

C. 21

D. 108

895. Right triangles: hypotenuse · Hard · MC

A right triangle has legs of length 3 and 4. What is the length of the hypotenuse?

A. 5

B. 7

C. 1

D. 12

896. Right triangles: hypotenuse · Hard · SPR

A right triangle has legs of length 6 and 8. What is the length of the hypotenuse?

897. Special right triangles: 45-45-90 · Medium · MC

An isosceles right triangle has a hypotenuse of length $6\sqrt{2}$. What is the length of each leg?

A. 3

B. 36

C. 12

D. 6

898. Special right triangles: 45-45-90 · Medium · MC

An isosceles right triangle has a hypotenuse of length $11\sqrt{2}$. What is the length of each leg?

A. 121

B. $\frac{11}{2}$

C. 22

D. 11

899. Special right triangles: 45-45-90 · Medium · MC

An isosceles right triangle has a hypotenuse of length $16\sqrt{2}$. What is the length of each leg?

A. 16

B. 32

C. 8

D. 256

900. Special right triangles: 45-45-90 · Medium · MC

An isosceles right triangle has a hypotenuse of length $17\sqrt{2}$. What is the length of each leg?

A. 34

B. 289

C. 17

D. $\frac{17}{2}$

901. Special right triangles: 45-45-90 · Medium · SPR

An isosceles right triangle has a hypotenuse of length $5\sqrt{2}$. What is the length of each leg?

902. Special right triangles: 45-45-90 · Medium · MC

An isosceles right triangle has a hypotenuse of length $7\sqrt{2}$. What is the length of each leg?

A. 7

B. $\frac{7}{2}$

C. 49

D. 14

903. Special right triangles: 45-45-90 · Medium · MC

An isosceles right triangle has a hypotenuse of length $18\sqrt{2}$. What is the length of each leg?

A. 324

B. 9

C. 18

D. 36

904. Special right triangles: 45-45-90 · Medium · SPR

An isosceles right triangle has a hypotenuse of length $3\sqrt{2}$. What is the length of each leg?

905. Special right triangles: 30-60-90 · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 14. What is the length of the hypotenuse?

A. 196

B. 28

C. 42

D. 14

906. Special right triangles: 30-60-90 · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 4. What is the length of the hypotenuse?

A. 12

B. 8

C. 4

D. 16

907. Special right triangles: 30-60-90 · Medium · SPR

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 13. What is the length of the hypotenuse?

908. Special right triangles: 30-60-90 · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 9. What is the length of the hypotenuse?

A. 81

B. 9

C. 27

D. 18

909. Special right triangles: 30-60-90 · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 7. What is the length of the hypotenuse?

A. 49

B. 21

C. 14

D. 7

910. Special right triangles: 30-60-90 · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 11. What is the length of the hypotenuse?

A. 22

B. 11

C. 121

D. 33

911. Area: rectangle · Easy · SPR

A rectangle has length 9 and width 6. What is its area?

912. Area: rectangle · Easy · MC

A rectangle has length 13 and width 11. What is its area?

A. 286

B. 24

C. 48

D. 143

913. Area: rectangle · Easy · MC

A rectangle has length 8 and width 4. What is its area?

A. 64

B. 24

C. 32

D. 12

914. Area: rectangle · Easy · SPR

A rectangle has length 12 and width 8. What is its area?

915. Area: rectangle · Easy · MC

A rectangle has length 9 and width 3. What is its area?

A. 54

B. 27

C. 12

D. 24

916. Area: rectangle · Easy · MC

A rectangle has length 8 and width 10. What is its area?

A. 18

B. 36

C. 160

D. 80

917. Area: triangle · Easy · MC
A triangle has a base of length 4 and a corresponding height of 10. What is the area of the triangle?
A. 10
B. 14
C. 40
D. 20

918. Area: triangle · Easy · SPR
A triangle has a base of length 5 and a corresponding height of 12. What is the area of the triangle?

919. Area: triangle · Easy · SPR
A triangle has a base of length 6 and a corresponding height of 12. What is the area of the triangle?

920. Area: triangle · Easy · MC
A triangle has a base of length 13 and a corresponding height of 12. What is the area of the triangle?
A. 39
B. 25
C. 156
D. 78

921. Area: triangle · Easy · SPR
A triangle has a base of length 11 and a corresponding height of 12. What is the area of the triangle?

922. Area: triangle · Easy · SPR
A triangle has a base of length 14 and a corresponding height of 4. What is the area of the triangle?

923. Area: triangle · Easy · MC
A triangle has a base of length 16 and a corresponding height of 10. What is the area of the triangle?
A. 80
B. 26
C. 40
D. 160

924. Area: triangle · Easy · MC
A triangle has a base of length 13 and a corresponding height of 4. What is the area of the triangle?
A. 52
B. 26
C. 13
D. 17

925. Volume: rectangular prism · Easy · MC
A rectangular prism has length 3, width 6, and height 4. What is the volume of the prism?
A. 13
B. 18
C. 72
D. 12

926. Volume: rectangular prism · Easy · MC

A rectangular prism has length 6, width 4, and height 2. What is the volume of the prism?

A. 48

B. 12

C. 24

D. 49

927. Volume: rectangular prism · Easy · MC

A rectangular prism has length 4, width 8, and height 4. What is the volume of the prism?

A. 32

B. 129

C. 16

D. 128

928. Volume: rectangular prism · Easy · MC

A rectangular prism has length 5, width 3, and height 4. What is the volume of the prism?

A. 12

B. 60

C. 15

D. 20

929. Volume: rectangular prism · Easy · MC

A rectangular prism has length 10, width 3, and height 7. What is the volume of the prism?

A. 210

B. 30

C. 70

D. 20

930. Volume: rectangular prism · Easy · SPR

A rectangular prism has length 6, width 3, and height 8. What is the volume of the prism?

931. Volume: rectangular prism · Easy · MC

A rectangular prism has length 8, width 6, and height 8. What is the volume of the prism?

A. 64

B. 48

C. 22

D. 384

932. Volume: rectangular prism · Easy · MC

A rectangular prism has length 5, width 6, and height 3. What is the volume of the prism?

A. 90

B. 14

C. 30

D. 15

933. Volume: cylinder, coefficient of pi · Hard · MC

A right circular cylinder has radius 4 and height 9. The volume is $k\pi$. What is the value of k?

A. 36

B. 72

C. 25

D. 144

934. Volume: cylinder, coefficient of pi · Hard · MC

A right circular cylinder has radius 5 and height 5. The volume is $k\pi$. What is the value of k?

A. 25

B. 30

C. 125

D. 50

935. Volume: cylinder, coefficient of pi · Hard · SPR

A right circular cylinder has radius 4 and height 7. The volume is $k\pi$. What is the value of k?

936. Volume: cylinder, coefficient of pi · Hard · SPR

A right circular cylinder has radius 3 and height 10. The volume is $k\pi$. What is the value of k?

937. Volume: cylinder, coefficient of pi · Hard · MC

A right circular cylinder has radius 3 and height 5. The volume is $k\pi$. What is the value of k?

A. 45

B. 14

C. 30

D. 15

938. Volume: cylinder, coefficient of pi · Hard · MC

A right circular cylinder has radius 2 and height 11. The volume is $k\pi$. What is the value of k?

A. 22

B. 44

C. 88

D. 15

939. Circles: circumference coefficient · Easy · MC

A circle has radius 3. The circumference is $k\pi$. What is the value of k?

A. 6

B. 5

C. 3

D. 9

940. Circles: circumference coefficient · Easy · MC

A circle has radius 10. The circumference is $k\pi$. What is the value of k?

A. 10

B. 100

C. 20

D. 12

941. Circles: circumference coefficient · Easy · SPR

A circle has radius 4. The circumference is $k\pi$. What is the value of k?

942. Circles: circumference coefficient · Easy · MC

A circle has radius 9. The circumference is $k\pi$. What is the value of k?

A. 18

B. 81

C. 9

D. 11

943. Circles: circumference coefficient · Easy · MC

A circle has radius 12. The circumference is $k\pi$. What is the value of k?

A. 24

B. 12

C. 144

D. 14

944. Circles: circumference coefficient · Easy · MC

A circle has radius 7. The circumference is $k\pi$. What is the value of k?

A. 9

B. 14

C. 7

D. 49

945. Circles: radius from the equation · Medium · MC

What is the radius of the circle with equation $(x-2)^2+(y-1)^2=9$?

A. 3

B. 2

C. 6

D. 9

946. Circles: radius from the equation · Medium · MC

What is the radius of the circle with equation $(x-3)^2+(y-3)^2=49$?

A. 3

B. 7

C. 14

D. 49

947. Circles: radius from the equation · Medium · MC

What is the radius of the circle with equation $(x-4)^2+(y-1)^2=25$?

A. 25

B. 5

C. 10

D. 4

948. Circles: radius from the equation · Medium · MC

What is the radius of the circle with equation $(x-5)^2+(y-6)^2=9$?

A. 5

B. 6

C. 9

D. 3

949. Circles: radius from the equation · Medium · MC
What is the radius of the circle with equation $(x-4)^2+(y-1)^2=16$?
A. 1
B. 4
C. 16
D. 8

950. Circles: radius from the equation · Medium · MC
What is the radius of the circle with equation $(x-5)^2+(y-2)^2=36$?
A. 6
B. 5
C. 36
D. 12

951. Circles: radius from the equation · Medium · MC
What is the radius of the circle with equation $(x-6)^2+(y-2)^2=25$?
A. 5
B. 10
C. 6
D. 25

952. Circles: radius from the equation · Medium · MC
What is the radius of the circle with equation $(x-6)^2+(y-4)^2=9$?
A. 9
B. 6
C. 4
D. 3

953. Circles: complete the square · Hard · MC
What is the radius of the circle with equation $x^2-6x+y^2+8y=-16$?
A. 4
B. 3
C. 9
D. 6

954. Circles: complete the square · Hard · SPR
What is the radius of the circle with equation $x^2-6x+y^2+8y=0$?

955. Circles: complete the square · Hard · MC
What is the radius of the circle with equation $x^2-10x+y^2+2y=10$?
A. 6
B. 5
C. 1
D. 36

956. Circles: complete the square · Hard · MC

What is the radius of the circle with equation $x^2 - 4x + y^2 + 2y = 11$?

A. 16

B. 1

C. 2

D. 4

957. Circles: complete the square · Hard · MC

What is the radius of the circle with equation $x^2 - 10x + y^2 + 4y = -4$?

A. 2

B. 5

C. 10

D. 25

958. Circles: complete the square · Hard · MC

What is the radius of the circle with equation $x^2 - 8x + y^2 + 2y = 19$?

A. 36

B. 4

C. 1

D. 6

959. Triangles: similarity scale · Medium · SPR

Two triangles are similar. A pair of corresponding sides measure 4 and 8. Another side of the smaller triangle measures 8. What is the length of the corresponding side of the larger triangle?

960. Triangles: similarity scale · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 7 and 14. Another side of the smaller triangle measures 11. What is the length of the corresponding side of the larger triangle?

A. 22

B. 77

C. $\frac{11}{2}$

D. 13

961. Triangles: similarity scale · Medium · SPR

Two triangles are similar. A pair of corresponding sides measure 6 and 12. Another side of the smaller triangle measures 11. What is the length of the corresponding side of the larger triangle?

962. Triangles: similarity scale · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 3 and 12. Another side of the smaller triangle measures 6. What is the length of the corresponding side of the larger triangle?

A. 24

B. 18

C. $\frac{3}{2}$

D. 10

963. Triangles: similarity scale · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 8 and 32. Another side of the smaller triangle measures 11. What is the length of the corresponding side of the larger triangle?

A. 44

B. $\frac{11}{4}$

C. 88

D. 15

964. Triangles: similarity scale · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 5 and 10. Another side of the smaller triangle measures 8. What is the length of the corresponding side of the larger triangle?

A. 10

B. 16

C. 40

D. 4

965. Triangles: similarity scale · Medium · SPR

Two triangles are similar. A pair of corresponding sides measure 4 and 8. Another side of the smaller triangle measures 7. What is the length of the corresponding side of the larger triangle?

966. Triangles: similarity scale · Medium · SPR

Two triangles are similar. A pair of corresponding sides measure 3 and 6. Another side of the smaller triangle measures 6. What is the length of the corresponding side of the larger triangle?

967. Right-triangle trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 9, the side adjacent to angle A has length 12, and the hypotenuse has length 15. What is the value of $\sin A$?

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. $\frac{4}{5}$

D. $\frac{3}{5}$

968. Right-triangle trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 5, the side adjacent to angle A has length 12, and the hypotenuse has length 13. What is the value of $\cos A$?

A. $\frac{5}{12}$

B. $\frac{12}{5}$

C. $\frac{12}{13}$

D. $\frac{13}{5}$

969. Right-triangle trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 3, the side adjacent to angle A has length 4, and the hypotenuse has length 5. What is the value of $\sin A$?

A. $\frac{3}{4}$

B. $\frac{3}{5}$

C. $\frac{4}{3}$

D. $\frac{4}{5}$

970. Right-triangle trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 7, the side adjacent to angle A has length 24, and the hypotenuse has length 25. What is the value of $\sin A$?

A. $\frac{24}{7}$

B. $\frac{7}{24}$

C. $\frac{24}{25}$

D. $\frac{7}{25}$

971. Right-triangle trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 16, and the hypotenuse has length 20. What is the value of $\sin A$?

A. $\frac{4}{3}$

B. $\frac{3}{4}$

C. $\frac{4}{5}$

D. $\frac{3}{5}$

972. Right-triangle trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 7, the side adjacent to angle A has length 24, and the hypotenuse has length 25. What is the value of $\cos A$?

A. $\frac{24}{25}$

B. $\frac{7}{24}$

C. $\frac{24}{7}$

D. $\frac{25}{7}$

973. Right-triangle trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 20, the side adjacent to angle A has length 21, and the hypotenuse has length 29. What is the value of $\tan A$?

A. $\frac{21}{29}$

B. $\frac{21}{20}$

C. $\frac{20}{21}$

D. $\frac{29}{20}$

974. Right-triangle trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 16, and the hypotenuse has length 20. What is the value of $\cos A$?

A. $\frac{4}{3}$

B. $\frac{5}{3}$

C. $\frac{4}{5}$

D. $\frac{3}{4}$

975. Circles: arc length coefficient · Medium · MC

A circle has radius 12. What is the length of a 120° arc of the circle, expressed as $k\pi$? Enter the value of k.

A. 12

B. $\frac{1}{3}$

C. 8

D. 4

976. Circles: arc length coefficient · Medium · MC

A circle has radius 9. What is the length of a 60° arc of the circle, expressed as $k\pi$? Enter the value of k.

A. 3

B. $\frac{1}{6}$

C. 9

D. $\frac{3}{2}$

977. Circles: arc length coefficient · Medium · SPR

A circle has radius 9. What is the length of a 90° arc of the circle, expressed as $k\pi$? Enter the value of k.

978. Circles: arc length coefficient · Medium · MC

A circle has radius 6. What is the length of a 90° arc of the circle, expressed as $k\pi$? Enter the value of k.

A. $\frac{3}{2}$

B. $\frac{1}{4}$

C. 6

D. 3

979. Circles: arc length coefficient · Medium · MC

A circle has radius 6. What is the length of a 120° arc of the circle, expressed as $k\pi$? Enter the value of k.

A. 4

B. $\frac{1}{3}$

C. 6

D. 2

980. Circles: arc length coefficient · Medium · MC

A circle has radius 6. What is the length of a 60° arc of the circle, expressed as $k\pi$? Enter the value of k.

A. 6

B. 2

C. $\frac{1}{6}$

D. 1

981. Circles: arc length coefficient · Medium · SPR

A circle has radius 9. What is the length of a 120° arc of the circle, expressed as $k\pi$? Enter the value of k.

982. Circles: arc length coefficient · Medium · MC

A circle has radius 12. What is the length of a 90° arc of the circle, expressed as $k\pi$? Enter the value of k.

A. 12

B. 6

C. 3

D. $\frac{1}{4}$

983. Circles: inscribed angle · Hard · MC

A central angle measures 80 degrees and intercepts an arc. An inscribed angle intercepts the same arc. What is the measure, in degrees, of the inscribed angle?

A. 80

B. 160

C. 100

D. 40

984. Circles: inscribed angle · Hard · MC

A central angle measures 40 degrees and intercepts an arc. An inscribed angle intercepts the same arc. What is the measure, in degrees, of the inscribed angle?

A. 140

B. 80

C. 20

D. 40

985. Circles: inscribed angle · Hard · MC

A central angle measures 60 degrees and intercepts an arc. An inscribed angle intercepts the same arc. What is the measure, in degrees, of the inscribed angle?

A. 60

B. 30

C. 120

D. 90

986. Circles: inscribed angle · Hard · MC

A central angle measures 100 degrees and intercepts an arc. An inscribed angle intercepts the same arc. What is the measure, in degrees, of the inscribed angle?

A. 80

B. 100

C. 50

D. 200

987. Circles: inscribed angle · Hard · MC

A central angle measures 120 degrees and intercepts an arc. An inscribed angle intercepts the same arc. What is the measure, in degrees, of the inscribed angle?

A. 90

B. 240

C. 60

D. 120

988. Circles: inscribed angle · Hard · MC

A central angle measures 50 degrees and intercepts an arc. An inscribed angle intercepts the same arc. What is the measure, in degrees, of the inscribed angle?

A. 130

B. 25

C. 50

D. 100

989. Volume: scale factor · Hard · MC

Every length of a rectangular prism is multiplied by 3. The original volume is 8. What is the volume of the new prism?

A. 24

B. 72

C. 11

D. 216

990. Volume: scale factor · Hard · SPR

Every length of a rectangular prism is multiplied by 3. The original volume is 12. What is the volume of the new prism?

991. Volume: scale factor · Hard · MC

Every length of a rectangular prism is multiplied by 2. The original volume is 5. What is the volume of the new prism?

A. 40

B. 20

C. 10

D. 7

992. Volume: scale factor · Hard · MC

Every length of a rectangular prism is multiplied by 3. The original volume is 10. What is the volume of the new prism?

A. 13

B. 30

C. 90

D. 270

993. Volume: scale factor · Hard · MC

Every length of a rectangular prism is multiplied by 2. The original volume is 10. What is the volume of the new prism?

A. 80

B. 40

C. 20

D. 12

994. Volume: scale factor · Hard · MC

Every length of a rectangular prism is multiplied by 2. The original volume is 8. What is the volume of the new prism?

A. 32

B. 10

C. 64

D. 16

995. Trigonometry: cofunction of the other acute angle · Hard · MC

In a right triangle, $\sin A = 7/25$. Angle B is the other acute angle. What is $\cos B$?

A. $\frac{7}{25}$

B. $\frac{18}{25}$

C. $\frac{7}{18}$

D. $\frac{25}{7}$

996. Trigonometry: cofunction of the other acute angle · Hard · MC

In a right triangle, $\sin A = 5/13$. Angle B is the other acute angle. What is $\cos B$?

A. $\frac{5}{8}$

B. $\frac{5}{13}$

C. $\frac{8}{13}$

D. $\frac{13}{5}$

997. Trigonometry: cofunction of the other acute angle · Hard · MC

In a right triangle, $\sin A = 8/17$. Angle B is the other acute angle. What is $\cos B$?

A. $\frac{9}{17}$

B. $\frac{8}{9}$

C. $\frac{17}{8}$

D. $\frac{8}{17}$

998. Trigonometry: cofunction of the other acute angle · Hard · MC

In a right triangle, $\sin A = 3/5$. Angle B is the other acute angle. What is $\cos B$?

A. $\frac{3}{2}$

B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. $\frac{5}{3}$

999. Trigonometry: cofunction of the other acute angle · Hard · SPR

In a right triangle, $\sin A = 9/41$. Angle B is the other acute angle. What is $\cos B$?

1000. Trigonometry: cofunction of the other acute angle · Hard · SPR

In a right triangle, $\sin A = 4/5$. Angle B is the other acute angle. What is $\cos B$?

Answer key

Student-produced responses are printed as the number to enter. Multiple-choice answers are the letter.

1. A
2. A
3. 2
4. D
5. 3
6. B
7. A
8. D
9. B
10. A
11. A
12. B
13. 6
14. B
15. 14
16. 7
17. 10
18. A
19. B
20. B
21. D
22. -1
23. -4
24. D
25. A
26. B
27. 8
28. D
29. A
30. A
31. C
32. B
33. A
34. A
35. D
36. B
37. C
38. B
39. C
40. C
41. A
42. B
43. C
44. C
45. D
46. C
47. B
48. B
49. C
50. D
51. A
52. C
53. A
54. A
55. B
56. D
57. A
58. D
59. B
60. D
61. A
62. B
63. A
64. A
65. A
66. A
67. D
68. C
69. A
70. A
71. D
72. B
73. B
74. D
75. A
76. A
77. C
78. D
79. 1/3
80. B
81. B
82. A
83. A
84. D
85. A
86. C
87. B
88. 6
89. A
90. D
91. D
92. C
93. C
94. D
95. B
96. B
97. A
98. D
99. C
100. -2
101. 3
102. C
103. A
104. B
105. D
106. B
107. 4
108. 5
109. B
110. C
111. -3
112. -3
113. 1
114. A
115. A
116. -1
117. -2/3
118. C
119. C
120. -1/2
121. A
122. A
123. D
124. A
125. -1/4
126. C
127. D
128. A
129. D
130. A
131. C
132. B
133. C
134. C
135. D
136. A
137. D
138. D
139. D
140. C
141. D
142. A
143. C
144. D
145. A
146. A
147. A
148. B
149. B
150. C
151. D
152. D
153. A
154. B
155. B
156. A
157. D
158. B
159. B
160. A
161. D
162. C
163. D
164. A
165. C
166. B
167. C
168. B
169. D
170. D
171. B
172. C
173. D
174. D
175. A
176. C
177. A
178. A
179. C
180. B
181. B
182. B
183. A
184. C
185. B
186. A
187. C
188. A
189. C
190. C
191. D
192. C
193. A
194. D
195. B
196. B
197. D
198. A
199. A
200. A
201. B
202. D
203. B
204. A
205. A
206. C
207. C
208. D
209. D
210. B
211. C
212. D
213. 4
214. 4
215. D
216. A
217. A
218. -4
219. C
220. A
221. C
222. C
223. B
224. 0
225. D
226. B
227. C
228. 8
229. D
230. A
231. D
232. -1
233. C
234. A
235. 8
236. 4
237. A
238. B
239. D
240. D
241. B
242. B
243. B
244. 9
245. 16
246. -2
247. 7
248. B
249. 0
250. A
251. D
252. C
253. D
254. A
255. C

256. B
257. B
258. 11
259. 18
260. 17
261. B
262. C
263. C
264. A
265. 15
266. C
267. C
268. C
269. C
270. D
271. B
272. A
273. C
274. B
275. B
276. D
277. D
278. B
279. C
280. C
281. C
282. D
283. B
284. C
285. C
286. C
287. 44
288. A
289. C
290. C
291. A
292. 35
293. A
294. D
295. -11
296. A
297. 52
298. 52
299. B
300. C
301. B
302. 40
303. 68
304. B
305. B
306. C
307. B
308. C
309. 55
310. D
311. D
312. 7
313. A
314. B
315. B
316. C
317. A
318. 11
319. 13
320. 5
321. A
322. 8
323. A
324. 9
325. A
326. A
327. D
328. D
329. C
330. C
331. A
332. D
333. 11
334. A
335. D
336. 4
337. C
338. C
339. C
340. B
341. A
342. D
343. D
344. A
345. D
346. C
347. C
348. C
349. C
350. A
351. D
352. C
353. C
354. A
355. B
356. D
357. D
358. B
359. B
360. D
361. C
362. A
363. A
364. C
365. A
366. C
367. D
368. D
369. B
370. D
371. A
372. A
373. A
374. D
375. A
376. D
377. A
378. D
379. B
380. A
381. D
382. A
383. B
384. B
385. C
386. D
387. D
388. D
389. C
390. C
391. A
392. D
393. A
394. A
395. A
396. D
397. C
398. A
399. B
400. C
401. D
402. A
403. C
404. D
405. C
406. B
407. A
408. C
409. B
410. C
411. A
412. C
413. A
414. D
415. A
416. B
417. A
418. C
419. C
420. B
421. C
422. D
423. 9
424. -5
425. D
426. A
427. D
428. 14
429. 12
430. 7
431. A
432. D
433. D
434. 4
435. C
436. A
437. A
438. 4
439. A
440. D
441. C
442. A
443. C
444. -12
445. B
446. A
447. A
448. -24
449. C
450. 24
451. D
452. 88
453. B
454. C
455. D
456. 8
457. D
458. 2
459. 8
460. D
461. C
462. B
463. B
464. 7
465. 7
466. A
467. B
468. D
469. C
470. -5
471. 9
472. C
473. B
474. B
475. -1
476. D
477. A
478. A
479. 5
480. C
481. D
482. A
483. C
484. D
485. C
486. C
487. A
488. A
489. 9
490. 4
491. 9
492. C
493. D
494. B
495. C
496. D
497. A
498. C
499. A
500. A
501. C
502. D
503. C
504. A
505. A
506. D
507. 54
508. C
509. C
510. 1536
511. A
512. C
513. B
514. A
515. 112
516. C
517. 1280
518. 24
519. B
520. A
521. D
522. C
523. D
524. D
525. 6
526. 12
527. D
528. C
529. B
530. B
531. B
532. A

533. B
534. B
535. D
536. C
537. C
538. D
539. B
540. B
541. D
542. C
543. A
544. D
545. B
546. B
547. 15
548. B
549. 20
550. 10
551. 20
552. D
553. 8
554. C
555. D
556. 6
557. D
558. D
559. D
560. C
561. D
562. 8
563. D
564. B
565. C
566. D
567. 15
568. C
569. B
570. D
571. D
572. D
573. B
574. 16
575. C
576. D
577. A
578. 15
579. 49
580. -40
581. B
582. B
583. C
584. D
585. -3
586. D
587. A
588. 5
589. A
590. B
591. B
592. C
593. D
594. D
595. B
596. B
597. C
598. A
599. 2
600. 71
601. B
602. 24
603. D
604. C
605. D
606. C
607. D
608. 8
609. A
610. B
611. 7
612. D
613. B
614. C
615. D
616. 4
617. 6
618. B
619. D
620. A
621. 2
622. A
623. 14
624. D
625. C
626. 4
627. A
628. C
629. B
630. A
631. B
632. A
633. C
634. D
635. B
636. 16
637. C
638. C
639. 81
640. B
641. B
642. A
643. B
644. C
645. C
646. A
647. B
648. A
649. B
650. C
651. B
652. D
653. B
654. A
655. 24
656. 4
657. B
658. C
659. C
660. A
661. D
662. B
663. A
664. B
665. 12
666. A
667. 4
668. 22
669. A
670. -36
671. D
672. D
673. C
674. D
675. D
676. C
677. -30
678. C
679. -16
680. D
681. C
682. D
683. D
684. A
685. D
686. C
687. D
688. B
689. B
690. A
691. B
692. C
693. B
694. D
695. D
696. A
697. A
698. B
699. D
700. C
701. C
702. A
703. 25
704. D
705. A
706. 14
707. C
708. D
709. 20
710. 30
711. D
712. B
713. 59
714. D
715. 43
716. B
717. A
718. A
719. 36
720. B
721. 24
722. C
723. C
724. 15
725. B
726. 28
727. D
728. A
729. C
730. C
731. D
732. B
733. 4
734. 18
735. 72
736. A
737. 70
738. B
739. D
740. D
741. B
742. D
743. C
744. 28
745. B
746. 150
747. B
748. C
749. B
750. C
751. 144
752. D
753. 48
754. A
755. A
756. B
757. 162
758. 108
759. B
760. D
761. D
762. A
763. B
764. D
765. 11
766. D
767. B
768. B
769. A
770. C
771. 14
772. 16
773. 8
774. C
775. D
776. 21
777. D
778. A
779. B
780. A
781. D
782. D
783. D
784. B
785. 2/7
786. A
787. 3/5
788. B
789. C
790. C
791. C
792. B
793. D
794. A
795. A
796. 5/7
797. B
798. A
799. B
800. 1/2
801. B
802. D
803. 3/4
804. B
805. 13/18
806. A
807. D
808. C
809. B

810. D
811. B
812. C
813. 47
814. D
815. D
816. D
817. B
818. B
819. D
820. B
821. A
822. A
823. B
824. A
825. D
826. A
827. B
828. C
829. D
830. D
831. C
832. C
833. B
834. B
835. B
836. B
837. B
838. A
839. B
840. C
841. D
842. C
843. B
844. C
845. B
846. B
847. B
848. B
849. C
850. B
851. 82
852. A
853. A
854. D
855. A
856. 73
857. D
858. D
859. D
860. A
861. C
862. A
863. C
864. 117
865. D
866. B
867. 63
868. 69
869. 91
870. C
871. 141
872. 125
873. 130
874. 69
875. B
876. 107
877. D
878. B
879. 68
880. D
881. C
882. B
883. B
884. D
885. B
886. A
887. C
888. 35
889. 24
890. A
891. B
892. B
893. 13
894. A
895. A
896. 10
897. D
898. D
899. A
900. C
901. 5
902. A
903. C
904. 3
905. B
906. B
907. 26
908. D
909. C
910. A
911. 54
912. D
913. C
914. 96
915. B
916. D
917. D
918. 30
919. 36
920. D
921. 66
922. 28
923. A
924. B
925. C
926. A
927. D
928. B
929. A
930. 144
931. D
932. A
933. D
934. C
935. 112
936. 90
937. A
938. B
939. A
940. C
941. 8
942. A
943. A
944. B
945. A
946. B
947. B
948. D
949. B
950. A
951. A
952. D
953. B
954. 5
955. A
956. D
957. B
958. D
959. 16
960. A
961. 22
962. A
963. A
964. B
965. 14
966. 12
967. D
968. C
969. B
970. D
971. D
972. A
973. C
974. C
975. C
976. A
977. 9/2
978. D
979. A
980. B
981. 6
982. B
983. D
984. C
985. B
986. C
987. C
988. B
989. D
990. 324
991. A
992. D
993. A
994. C
995. A
996. B
997. D
998. C
999. 9/41
1000. 4/5